International Journal of Digital Earth

ISSN: 1753-8947 (Print) 1753-8955 (Online) Journal homepage: www.tandfonline.com/journals/tjde20

An adaptive Segment Anything Model 2 (SAM2) for planted field segmentation from remote sensing imagery

Shuaijun Liu , Qi Dong , Xuehong Chen , Xiuchun Dong , Ping Huang , Peng Yang & Jin Chen

To cite this article: Shuaijun Liu , Qi Dong , Xuehong Chen , Xiuchun Dong , Ping Huang , Peng Yang & Jin Chen (2026) An adaptive Segment Anything Model 2 (SAM2) for planted field segmentation from remote sensing imagery, International Journal of Digital Earth, 19:1, 2645885, DOI: 10.1080/17538947.2026.2645885

To link to this article: https://doi.org/10.1080/17538947.2026.2645885

Published online: 21 Mar 2026.

INTERNATIONAL JOURNAL OF DIGITAL EARTH
2026, VOL. 19, NO. 1, 2645885
https://doi.org/10.1080/17538947.2026.2645885

RESEARCH ARTICLE

OPEN ACCESS Check for updates

An adaptive Segment Anything Model 2 (SAM2) for planted field segmentation from remote sensing imagery

Shuaijun Liu[a,b], Qi Dong[b], Xuehong Chen[b], Xiuchun Dong[c], Ping Huang[c], Peng Yang[d] and Jin Chen[b]

[a]Jiangsu Key Laboratory of Atmospheric Environment Monitoring and Pollution Control, Collaborative In-novation Center of Atmospheric Environment and Equipment Technology, School of Environmental Sci-ence and Engineering, Nanjing University of Information Science and Technology, Nanjing, Jiangsu, People's Republic of China; [b]State Key Laboratory of Earth Surface Processes and Resource Ecology, Institute of Remote Sensing Science and Engineering, Faculty of Geographical Science, Beijing Normal University, Beijing, People's Republic of China; [c]State Key Institute of Remote Sensing and Digital Agriculture, Sichuan Academy of Agricultural Science, Chengdu, People's Republic of China; [d]Institute of Agricultural Resources and Regional Planning, Chinese Academy of Agricultural Sciences, Beijing, People's Republic of China

ABSTRACT

Planted fields, which are defined by areas containing different crop types, are essential for agricultural planning, crop mapping, and yield estimation. Accurate and timely information on planted fields is crucial. The Segment Anything Model 2 (SAM2) offers advanced image segmentation but is not directly suited for planted field segmentation. We propose ASAMPS, an Adaptive SAM2 model tailored for segmenting planted fields from single-date or time series remote sensing images. Unlike other SAM-based adaptations that require fine-tuning the model, ASAMPS incorporates three external modules to automatically generate and optimize prompt points, aligning SAM2 with the specific requirements of planted field segmentation without any retraining. Testing across six agricultural regions in China and the United States, ASAMPS demonstrated robust performance, achieving accuracy comparable to or surpassing state-of-the-art supervised methods. It demonstrated strong performance in complex landscapes and was effective across multiple satellite platforms, including Planet, GF-2, Sentinel-2, and Landsat 8 OLI. Additionally, ASAMPS efficiently extracted minimal planted fields from multi-temporal imagery. ASAMPS leverages SAM2's capabilities without requiring retraining and offers enhanced flexibility for optimizing prompts, making it suitable for diverse agricultural monitoring applications.

ARTICLE HISTORY
Received 26 November 2024
Accepted 11 March 2026

KEYWORDS
Segment Anything Model 2 (SAM2); planted field extraction; image segmentation; prompts; remote sensing imagery

1. Introduction

An agricultural field, alternatively referred to as a cropland parcel, plot, or patch, or simply as farmland, constitutes the fundamental spatial unit for agricultural practice. Agricultural fields can generally be divided into two distinct categories in the real world (as shown in Figure 1): Individual agricultural fields, which are demarcated by tangible, hard boundaries such as ridges, roads, and ditches (Yan and Roy 2014; Persello et al. 2019; Liu et al. 2022); and planted fields, which are spatially separated by the cultivation of different crop types (soft edges) rather than hard edges. Individual agricultural fields are characteristically stable over the years. Their boundaries are often directly linked to land ownership through the presence of hard edges. Planted fields, in contrast, represent temporary spatial units that vary with the different types of crops cultivated over different years or seasons. As a result, multiple planted fields may exist within an individual agricultural field. Since planted fields are the basic spatial unit in which all cropping activities take place during a growing season, accurate and timely information on planted fields has become a prerequisite for planning cooperative agricultural activities such as irrigation, fertilisation, and harvesting. Moreover, it is a fundamental unit for crop mapping and yield estimation (Matton et al. 2015; Waldner and Diakogiannis 2020). In this context, with the increasing availability of high-resolution satellite

CONTACT Jin Chen chenjin@bnu.edu.cn State Key Laboratory of Earth Surface Processes and Resource Ecology, Institute of Remote Sensing Science and Engineering, Faculty of Geographical Science, Beijing Normal University, Beijing, 100875, People's Republic of China
Supplemental data for this article can be accessed online at https://doi.org/10.1080/17538947.2026.2645885.

Figure 1. (a) Individual agricultural fields with hard edges and (b) planted fields spatially separated by the cultivation of different crop types (soft edges).

imagery, the automatic extraction of spatial details of planted fields in each growing season from high-resolution remotely sensed imagery has gained increasing attention worldwide.

Considering the pivotal role of agricultural fields in agricultural practice, a large number of remote sensing-based methods have been developed during the past decades as a typical application of image segmentation techniques. These methods predominantly focus on the automatic extraction of individual agricultural fields, known for their typically stable hard edges. Broadly, these methods can be divided into two types based on whether or not they use deep learning techniques: traditional and deep-learning-based methods. Traditional methods are primarily based on edge and/or region detection, utilising convolutional kernels as edge detectors, such as the Canny operator, χ algorithm, and Gaussian edge filter, to identify hard edges like ridges, paths, and ditches (Liu et al. 2010; Xi and Zhang 2012; Verrelst et al. 2014; Belgiu and Csillik 2018). The homogeneity of spectral or textural features within individual agricultural fields is then used to identify the spatial extent of individual agricultural fields through one of the classification, clustering, and watershed algorithms (Segl and Kaufmann 2001; Da Costa et al. 2007; Belgiu and Drăguţ 2014; García-Pedrero, Gonzalo-Martín, and Lillo-Saavedra 2017; Lebourgeois et al. 2017). Due to the lack of automated feature extraction capabilities, traditional methods, constrained by their dependence on manually defined spectral, textual, and edge features, often suffer from inconsistencies across users and study areas, reducing the robustness and accuracy of traditional methods (Aquino, Gegúndez-Arias, and Marín 2010). As a result, deep-learning-based methods are gradually replacing traditional methods due to their powerful feature mining and fitting capabilities. In general, deep-learning-based methods mainly include two types of methods: semantic segmentation and instance segmentation algorithms. Semantic segmentation aims to extract regions of different categories by classifying each pixel in the image into a corresponding category. In contrast, instance segmentation not only extracts regions of different categories but also distinguishes the specific object of the class, i.e. the instance. In agricultural field segmentation applications, semantic segmentation algorithms typically use a set of convolution operators to automatically extract spatial and spectral features (also known as dense features) at different spatial scales to accomplish the segmentation task (Long, Shelhamer, and Darrell 2015; Huang, Huang, and Krahenbuhl 2018; Ma et al. 2019; Persello et al. 2019; Gao, Liu, and Gong 2020; Meyer, Lemarchand, and Sidiropoulos 2020; Rabbi et al. 2020; Waldner and Diakogiannis 2020; Yang, Zhang, and Zhang 2020; Waldner et al. 2021; Liu et al. 2022; Tian et al. 2024). Such algorithms dominate the segmentation of individual agricultural fields due to the presence of hard edges. On the other hand, if each agricultural field is considered as an instance, then agricultural field segmentation is more like an instance segmentation task. Accordingly, many convolutional neural network-based instance segmentation algorithms, developed in other domains, such as Fully Convolutional Instance-aware Semantic Segmentation (FCIS) (Li et al. 2017), Mask R-CNN (He et al. 2017; Freitas et al. 2024), Cascade Mask R-CNN (Cai and Vasconcelos 2018), Mask Scoring R-CNN (Huang et al. 2019), and High-Quality Instance Segmentation Network (HQ-ISNet) (Su et al. 2020), have the potential to be used in the agricultural field segmentation task. Vision Transformers (ViT) have also shown great potential in instance segmentation by capturing long-range dependencies and global context, making them effective for complex tasks like agricultural field segmentation (Khan et al. 2025). However, their high computational cost and large training data requirements present challenges, especially with smaller training datasets. ViT adaptor models address some of these issues by introducing lightweight components that improve the efficiency of feature extraction and reduce memory consumption, making them more practical for more applications (Chen et al. 2022). While ViT Adaptor improves segmentation accuracy and training efficiency, it still faces challenges with higher computational demands and real-time processing. The main reason is that instance segmentation algorithms are difficult to train on relatively small datasets and are prone to overfitting. They often require more training data to outperform semantic segmentation algorithms (Liu et al. 2024).

Despite deep learning methods reducing the need for expert knowledge and enhancing segmentation accuracy through their automated feature extraction capabilities, as data-driven methods, these methods not only require a large amount of training data to be labelled, but also rely heavily on the training data to extract dense features across different spatial scales. As a result, the practical challenge is how to effectively label large amounts of training data and ensure improved transferability of deep learning models across different study areas or images (Yoo et al. 2016).

In comparison to individual agricultural fields, planted fields present a distinct set of characteristics. Firstly, the boundaries of planted fields may exhibit either hard or soft edges, attributable to spectral or textural variations arising from the cultivation of diverse crops in individual agricultural fields. Second, within a singular planted field, spectral or textural features tend to be more homogeneous due to growing the same crop. In contrast, an individual agricultural field might host a variety of crops, leading to reduced homogeneity in these features. Third, planted fields are transient spatial entities that vary seasonally and can be extracted either from a single image acquired at a single time or from multi-temporal images within a growing season. The use of multi-temporal images within a growing season better ensures that satellite sensors observe spectral or textural differences between crops within a growing season. Individual agricultural fields, on the other hand, tend to remain stable over the years due to the presence of invariant hard edges that allow their extraction from an image taken at any time during a year. Given the complex nature of planted fields, although many individual agricultural field methods have the potential to be used to extract planted fields, automated extraction of planted fields has not been fully explored and there are a few sophisticated methods specifically designed to extract planted fields in a growing season.

Recently, Meta AI Research released a revolutionary solution for image segmentation, suitable for both semantic and instance segmentation tasks: The Segment Anything Model (SAM) (Kirillov et al. 2023), which has significantly inspired a large amount of image segmentation studies tailored to specific applications. The exceptional performance of SAM can be attributed to three key factors: 1) a state-of-the-art Vision Transformer (ViT) with an attention mechanism, which ensures the capture of local and global features as well as spatial contextual dependencies from the image. 2) The largest training dataset to date (SA-1B), including 11 million diverse, high-resolution images and 1.1 billion segmentation masks, which ensures the generalisability of the model, i.e. SAM's performance on zero-shot (i.e. without additional sample training) transfer tasks is often comparable to, or even superior to, task-specific training models. 3) The ability to handle various types of prompts, including points, boxes, or text, which allows the model to handle a wide range of user-defined image segmentation tasks. Here, prompts are defined as user-defined input cues that guide the model to focus on specific parts of the image for segmentation. These prompts allow the user to specify which objects or areas the model should segment.

Building on the success of SAM, Segment Anything Model 2 (SAM2) introduces three key enhancements without changing the overall structure of SAM (Ravi et al. 2024). First, SAM2 improves SAM accuracy by incorporating more advanced transformer architectures that refine the ability to capture fine-grained detail and spatial relationships. Second, SAM2 improves prompt handling by introducing better prompt interpretability, allowing for more precise segmentation guidance in complex scenes. Third, SAM2 also reduces the computational load by using streaming memory and integrating a model-in-the-loop data engine, making it more efficient for real-time applications such as video segmentation. With the above improvements, SAM2 is expected to be applied to the task of agricultural field segmentation, especially to the segmentation of planted fields. Planted fields have greater spectral and textural homogeneity than individual agricultural fields, which are more similar to the segmentation objects defined in SAM2. Meanwhile, the problems of existing models that require labelling a large amount of training data and lack of spatio-temporal transferability could be effectively mitigated by the application of SAM2.

Significant efforts are underway within the remote sensing community to adapt SAM for domain-specific applications. For example, Ren et al. (2023) tested the zero-shot performance of SAM on five segmentation tasks, i.e. segmentation of solar panels, buildings, roads, and clouds using five very high resolution (VHR) images (sub-metre) and delineation of agricultural parcel boundaries (individual agricultural field) using a medium resolution image (10 metres). Ji et al. (2023) also investigated SAM on several remotely sensed image segmentation tasks, i.e. crop, building, and road segmentation. The above results show that the direct application of SAM to remotely sensed data is still unsatisfactory. Accordingly, several adaptation techniques have been developed to adapt SAM to specific remote sensing applications (Osco et al. 2023; Zhang et al. 2023; Chen et al. 2023a, 2023b). For example, Chen et al. (2023a) proposed a SAM adaptor to improve SAM performance in remote sensing imagery. Osco et al. (2023) tested a new method, called Text PerSAM-F, by combining text prompt with SAM to improve the generality of SAM. Chen et al. (2023b) proposed RSPrompter. Using prompt learning, RSPrompter learned the semantic and positional features from the intermediate feature layers of SAM backbone, and then generated optimal semantically and positionally matching prompts for SAM. In this way, RSPrompter-guided SAM achieved improved segmentation results. Similarly, Zhang et al. (2023) proposed the TEXT2SEG method with the function of generating semantically relevant prompts for SAM, which consequently enabled SAM to produce semantic segmentation results from a remotely sensed image. On the other hand, as a newly released model, SAM2 has yet to undergo extensive testing and validation in various real-world applications.

The unsatisfactory performance of direct application of SAM or SAM2 to remotely sensed data is due to the following reasons. The first reason is that SAM and SAM2 were trained on natural (RGB) images rather than remote sensing imagery. The natural images, which usually capture only the red, green, and blue colour bands, while remote sensing imagery usually contain a wider range of spectral bands (e.g. near-infra-red, red-edge, and short-wave infra-red bands). Therefore, the effective selection and utilisation of the rich and useful spectral information in remote sensing imagery becomes a critical issue for the direct application of SAM and SAM2 to remote sensing segmentation tasks. The second reason is the disparity in spatial resolution between natural and remote sensing imagery. Unlike natural images, remote sensing imagery has different spatial resolutions. This disparity affects SAM's ability to accurately segment objects of interest in remote sensing imagery with different spatial resolutions, because SAM and SAM2 were primarily trained on high-resolution natural images. In addition, the performance of SAM and SAM2 is highly dependent on the quality of the prompts. If the prompts provided are inaccurate or ambiguous, the model's segmentation results may be affected, particularly with respect to the position and number of prompts (Chen et al. 2023b). In the intensive task of remotely sensed image segmentation, manually selecting prompts may be impractical. Therefore, most recent studies have adapted SAM by automatically generating semantic prompts using extended models (Ji et al. 2023; Osco et al. 2023; Zhang et al. 2023; Chen et al. 2023b), although the results remain unsatisfactory. It is crucial to generate accurate prompts based on the characteristics of the planted fields, while maintaining the generalisation ability and operational efficiency of SAM and SAM2.

Based on the pros and cons of SAM series models, we present ASAMPS, an Adaptive SAM2 model for Planted field Segmentation. The core innovation of our method lies in its non-invasive strategy: instead of modifying SAM2's internal parameters, ASAMPS operates through external modules that automatically generate and optimise prompt points based on the unique characteristics of planted fields and remotely sensed imagery. This approach preserves SAM2's original architecture and powerful zero-shot generalisation capability in full, while enabling it to perform the specific, dense segmentation task of delineating planted fields. We tested ASAMPS at six experimental sites and compared it with a state-of-the-art semantic segmentation method for agricultural field segmentation, IAFRes (Waldner and Diakogiannis 2020), and a typical instance segmentation algorithm, ViT Adaptor (He et al. 2017; Freitas et al. 2024). The main objective of these experiments is to investigate whether ASAMPS can achieve better performance for the intensive task of planted field segmentation in the case of zero shots over different study areas or different spatial resolution images.

2. Methodology

2.1. Segment Anything Model 2 (SAM2)

SAM2 is an enhanced version of SAM that retains the core architecture of SAM (Kirillov et al. 2023), but introduces three major enhancements (see Introduction section) (Ravi et al. 2024). The SAM series models have demonstrated remarkable capabilities in natural image segmentation. The SAM series models consist of three main modules (Figure 2): image encoding, prompt encoding, and the decoding and mask generation. The process of image segmentation is as follows: First, the SAM series models use a pre-trained Vision Transformer (ViT) to process the input image and generate an embedded representation of the image in the image encoding process. This embedding is a compressed and abstracted representation of the input image, providing foundational information for subsequent mask generation. Prompt encoding is another critical component of the SAM series models. The module is capable of processing various types of prompts, including simple spatial information (points or boxes) and more complex mask prompts (text). These prompts are translated into a format that the SAM series models can understand, guiding it to specifically identify the extent of the segmentation area or objects in the image. Then, the mask decoder generates the final segmentation masks (e.g. planted field) by combining the image embedding and prompt encoding. Meanwhile, The SAM series models are able to generate a prediction score for each generated mask, suggesting that the SAM series models can not only identify the object in the image, but also evaluate the confidence of the prediction (for more detailed information about the scoring mechanism of the SAM series models, see Appendix I). This feature can help select the best segmentation result when ambiguous prompts may refer to multiple masks. This process can be expressed as:

$$Seg_{img} = V_{mask}(U_{img}(x),\ U_{pro}(p_{po\ int})) \tag{1}$$

where U_{img} denotes the image encoder, U_{pro} corresponds to the prompt encoder, and V_{mask} refers to the mask decoder. x denotes a remotely sensed image waiting to be processed, and P_{point} is the prompt information inputted by users. Seg_{img} corresponds to the predicted masks.

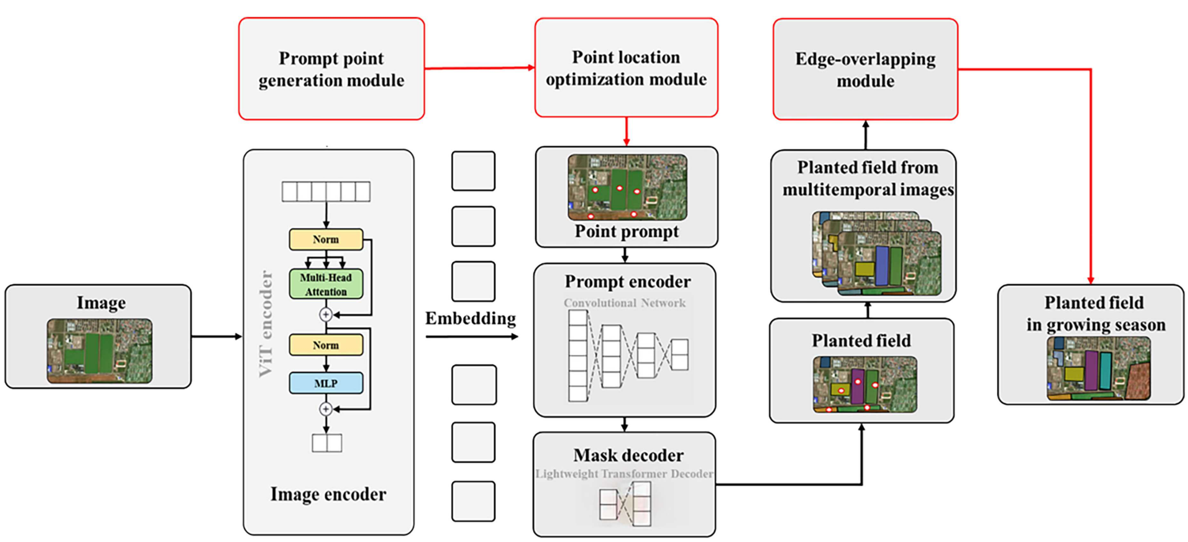

Figure 2. Framework of the SAM series models, where the red boxes and lines are the modules added by ASAMPS.

2.2. Motivation of developing ASAMPS

The user-defined prompts bridge the gap between natural image and user-defined segmentation tasks, distinguishing SAM2 from traditional image segmentation models. SAM2 uses prompts to directly guide the model to specify the extent of the segmentation area or objects for segmentation. This flexibility allows the user to dynamically adjust the segmentation results by changing the prompts, thereby increasing the usability and adaptability of the model to a wide range of image segmentation applications.

While prompts provide excellent adaptability for SAM2, there is often ambiguity in how to define these prompts, which poses a challenge for SAM2 in specialised applications. Since the point prompt is the most straightforward and can be automatically generated by SAM2, we used the point prompt as an example to investigate how the location and number of point prompts affect segmentation performance by varying the number and location of point prompts. Specifically, we used SAM2's grid sampling method to automatically generate different numbers of point prompts. The grid sampling method is an automatic method for generating point prompts that automatically divides the used image into multiple grids of user-defined size and uses the intersection points of the grids as point prompts. The more the grid is divided, the greater the number of point prompts (see Figure 3(a)). In addition, we used SAM2's point prompt input function to adjust the position of the point prompt, which allows the user to enter a point prompt at any position on the image (see Figure 3(b)).

Figure 3. Effect of different prompt points (a) and a prompt point at different locations within the same planted field (b) on planted field extraction. Score is the prediction score output by SAM2. The images on the left are original remote sensing imagery with prompt points, and the images on the right are segmentation results under prompt point input.

As shown in Figure 3(a), it is observed that increasing the number of prompt points can reduce the ambiguity of the prompts, resulting in the segmentation of more objects. The location of the prompt points also plays a crucial role in determining the accuracy of the segmentation. As shown in Figure 3(b-1), if a prompt point is accurately placed at or near the centre of a target object, the model can correctly separate the target object. Conversely, if a prompt point is placed at the edge of a target object, the model may produce less accurate segmentation results, such as separating out a portion of the target object (Figure 3(b-2)) or separating out an object larger than the actual target object (Figure 3(b-3)).

These results show that the number and location of the prompt points have a significant impact on the performance of SAM2, highlighting the importance of automatically generating the appropriate number and correct location of the prompt points. This observation became the foundation for developing the ASAMPS model for planted field segmentation.

2.3. ASAMPS

ASAMPS model is a SAM2-based method for extracting planted fields (instance-based segmentation) from single- or multi-temporal images within a growing season. It is characterised by automatically generating the appropriate number and correct location of prompt points based on the unique characteristics of planted fields in remote sensing imagery. This new model includes three modules as shown in Figure 2. The prompt point generation module is designed to automatically generate the appropriate number of prompt points from a single image. The prompt point optimisation module is dedicated to reducing the redundant points and errors caused by inaccurate prompt point location. The edge-overlapping module is designed to extract planted fields from multi-temporal images by overlapping the edges extracted from each individual image.

2.3.1. Prompt point generation module

The results shown in Figure 3 indicate that each planted field should have at least one prompt point located inside the planted field and away from the planted field edges. However, the edges of each planted field are unknown until the planted field segmentation is performed. Fortunately, regardless of whether the edges are hard or soft, the edges are located where the gradient values are large (Matton et al. 2015; Wagner and Oppelt 2020; Waldner and Diakogiannis 2020; Liu et al. 2022). Therefore, we can represent the approximate location of the edges by calculating the gradient values. Accordingly, the Sobel filter, an effective edge detection technique used to identify high-frequency parts in an image, is used to compute the gradient map (Equation 2). Then, the prompt points are placed at the locations where the gradient values are smaller, ensuring that the prompt points are away from the planted field edges.

$$Grad_{img} = sobel\,(x) \tag{2}$$

where x represents the input image of the study area, and $Grad_{\mathrm{img}}$ is the extracted gradient map.

In addition to the gradient value condition, another property of the planted field should be considered, namely, pixels inside the planted field should be more homogeneous in terms of spectral and textural features than pixels near the edges of the planted field. Following the work of Zhu et al. (2016), the category homogeneity index (HI) (Equation 3) is used here to calculate the category homogeneity for each pixel by comparing its category with neighbouring pixels. The category information is obtained using unsupervised classification algorithms such as ISODATA (Ball and Hall 1965) with the input of all spectral bands of the image.

$$\mathrm{HI}_i = \left(\sum_{k=1}^{m} I_k\right)/m \tag{3}$$

where I_{k} is the category uniformity value for each neighbouring pixel within a moving window of m pixels. When the k^{th} pixel has the same category value as the central pixel (i), the value is set to 1; otherwise, it is 0. The range of HI values is from 0 to 1, with higher values indicating more uniform category values.

By combining HI and gradient values, the prompt points can be generated at locations with larger HI values and smaller gradient values, ensuring that each prompt point is within the planted field and away from the planted field edges. The combination of HI and gradient can be implemented by a weighted index, Planted Index (PIE), defined in Equation 4.

$$PI_i = \varepsilon \cdot (1 - Grad_i) + (1 - \varepsilon) \cdot HI_i \tag{4}$$

where the variable ε represents a weight of gradient and (1- ε) represents a weight of HI. The value of ε can be dynamically adjusted according to the area of interest, we recommend 0.5. Before calculating *PIE*, the value of Grad_{img} for each pixel is normalised to a range from 0 to 1 using min-max normalisation. PI_i represents the potential of each pixel as a prompt point, with higher values indicating greater potential to be as a prompt. Consequently, we generate prompt points based on the magnitude of PI_i values, with the specific process as follows (Figure 4): first, the calculated PI_i values for each pixel are sorted from small to large, then a threshold δ is set based on experience or the predefined number of prompt points, with a default value of 0.85 in this study as discussed. The pixels with a PI_i value greater than the threshold are set as candidates for prompt points.

Since we focus only on agricultural land rather than other land cover types such as buildings, forests, water bodies, and roads, the existing land cover classification products, e.g. World Cover Dataset (Zanaga et al. 2022), Esri Land Cover Dataset (Karra et al. 2021), and Dynamic World Dataset (Brown et al. 2022), can be used to mask out the areas with other land cover types, leaving only agricultural land before calculating PI_i values. The Dynamic World Dataset, which provides near real-time land cover information at a 10-metre resolution, was used in this study. Its key advantage is the high temporal frequency, allowing the selection of a cloud-free, contemporaneous mask for each input image. However, as a globally produced automated classification product, it is not without errors. Misclassification, particularly the confusion between agricultural land and certain types of grasslands, bare soil, or urban areas, can occur. These errors would directly propagate to our results: erroneously omitted agricultural areas would not be processed by ASAMPS, while incorrectly included non-agricultural areas might generate false positive segmented fields. It is worth noting that the quality of these land cover datasets has a critical impact on the scope of implementation of planted field segmentation. Any classification errors in these land cover datasets could result in some areas of agricultural land not being segmented or areas of other land cover types being segmented. A comprehensive comparative analysis of the performance of ASAMPS when utilising different land cover products (e.g. Esri Land Cover, WorldCover) is beyond the scope of this study, as it would require an extensive evaluation of the accuracy of each product over our diverse test sites—a significant research question in itself. However, the choice of product is an important practical consideration. Products with

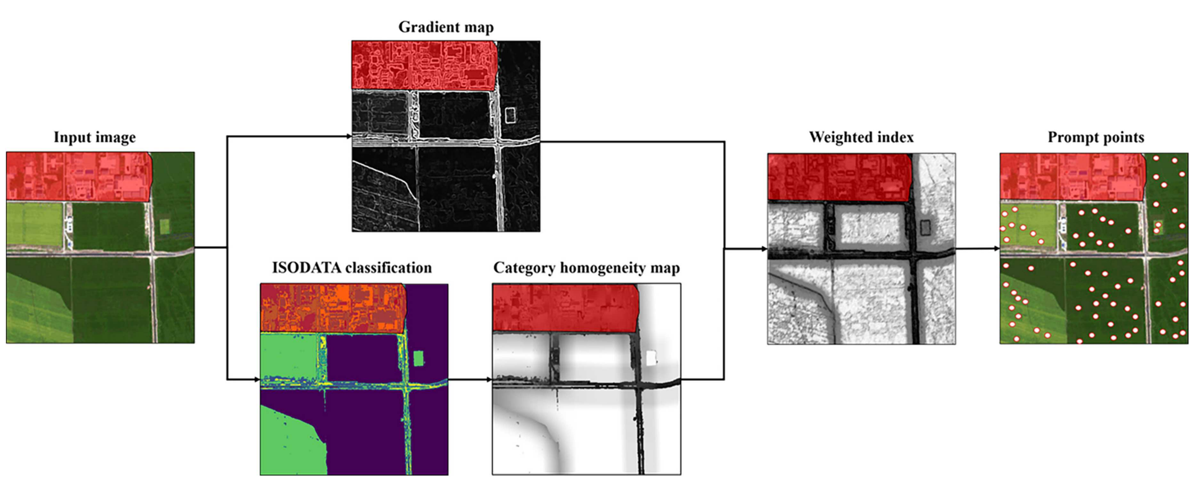

Figure 4. The process of automatic prompt point generation. The red area has been masked out as non-agricultural land areas prior to prompt point generation and excluded from the planted field segmentation.

higher spatial resolution (e.g. 10m for Dynamic World vs. 30m for Esri) generally allow for more precise boundary delineation of small fields. Conversely, products with higher thematic accuracy for the specific region of interest would minimise commission and omission errors in the initial mask. The framework of ASAMPS is agnostic to the specific land cover product used, and we recommend users select the most accurate and resolution-appropriate product available for their study area. We recommend that users verify and update these datasets before implementing the extraction process. To edit errors in existing land cover classification products, there are many commercial software programmes that provide interactive editing capabilities.

2.3.2. Point location optimisation module

In the previous step, we generated prompt points with a clear physical meaning, a large number of them, and some of them may be redundant and in the wrong locations. The main task of the optimisation module is to remove the redundant points and keep the better located prompt points. In SAM2 model, each extracted mask (e.g. planted field) receives a prediction score ranging from 0 to 1. This score is obtained for each segmented mask by averaging the likelihood of each pixel belonging to the segmented mask, and serves as an indicator of the confidence of the prediction (see Appendix I). This score is used to optimise the location of prompt points as the higher the score, the better the segmentation. Detailed information about the process is given below (Figure 5).

Based on our observations, when two prompt points are close to each other, the segmentation results will also be similar with close prediction score values, resulting in redundancy. To overcome the redundancy of adjacent prompt points, a dilation window is first created around each generated prompt

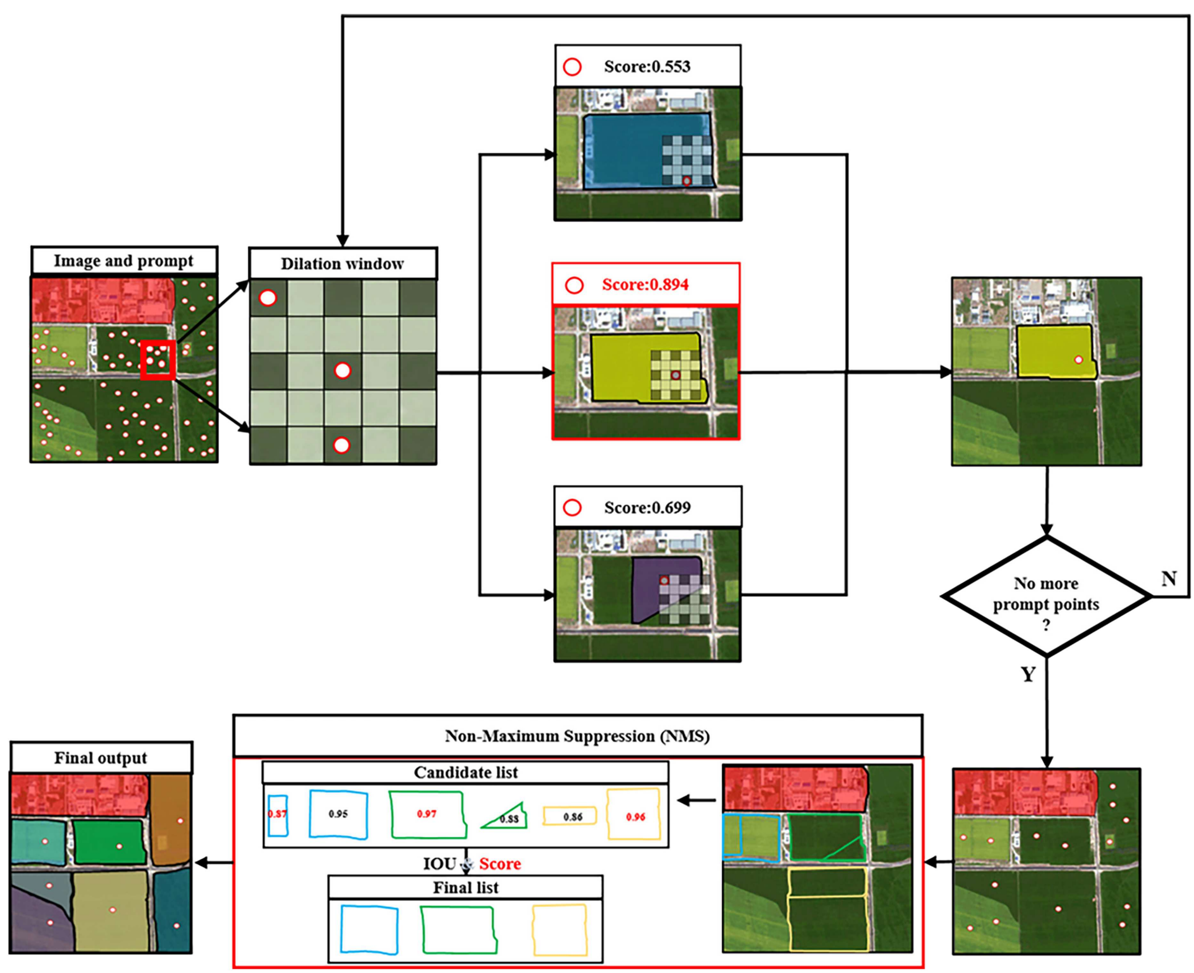

Figure 5. Point location optimisation process. The red area has been masked as non-agricultural areas.

point. Here, a dilation window refers to an expanded area around each prompt point like a moving window, but some of the pixels in the dilation window are skipped and not processed in the following flow. The dilation rate (denoted by *d*) determines how many pixels are skipped in a dilation window (Figure 5). The prompt points that are not skipped within the dilation window are selected. SAM2 then generates a mask for each of the selected prompt points and returns the prediction scores for them. The prompt point with the highest prediction score is selected and its location is retained from the selected prompt points within the dilation window (Figure 5). The above process is applied iteratively to all prompt points generated by the Prompt Point Generator module until no prompt point remains. Finally, only the prompt points that are selected at least once due to the highest prediction score in each dilation window are retained. Here, the use of a dilation window instead of a traditional window here has two advantages: first, it expands the search scope to include a more diverse set of prompt points, and second, it ensures that each selected prompt point maintains a certain spatial distance from the centre prompt point, which is conducive to reducing redundancy.

The above process does not limit the number of remaining prompt points within the same planted field, i.e. a planted field can have multiple remaining prompt points. It is necessary to further eliminate redundant prompt points to ensure that each planted field has only one remaining prompt point. Considering that the planted fields exhibit significant exclusivity in their spatial distribution, i.e. no two planted fields can spatially overlap, which means that none of the planted fields intersect other planted fields, Non-Maximum Suppression (NMS) (Rothe, Guillaumin, and Van Gool 2014) is then used to address this issue. In detail, first, all planted fields predicted by SAM2 with the input of the remaining prompt points are organised into a candidate list in descending order of prediction scores. The planted field with the highest score is selected from the candidate list. It is added to the final list and removed from the candidate list. NMS then computes the Intersection over Union (IoU) (Equation 5) between the highest scoring planted field and each of the remaining planted fields in the candidate list. The IoU metric quantifies the degree of overlap between two planted fields, i.e. an IoU value near 0 indicates minimal or no overlap. The overlapping planted field in the candidate list is removed from the candidate list if the IoU value of the planted field is greater than a threshold value (default value is 0.01). This step is critical because it ensures that only those planted fields that are spatially independent and do not overlap with the planted fields in the final list will remain in the candidate list. Then, the planted field with the highest score in the updated candidate list is added to the final list and removed from the candidate list. The above process is repeated iteratively until no planted fields remain in the candidate list. All remaining prompt points in the final list are then used as prompts for SAM2.

$$IOU(f_1, f_2) = \frac{area(f_1 \cap f_2)}{area(f_1 \cup f_2)} \tag{5}$$

where f_1 and f_2 denoted as two planted fields need to be analysed, *area* is the overlapping area between the two planted fields using the Intersection and Concatenation operators.

2.3.3. Edge-overlapping module

Ideally, single-date imagery would clearly show all planted field boundaries. In practice, however, images from different dates may show different planted field boundaries due to changing crop phenology, different planted and harvesting schedules, and varying soil moisture. Consequently, planted fields extracted from a single-date image may not accurately represent the actual planted fields in a growing season. The edge-overlapping module is designed to address the issue. We employ the edge-overlapping model inspired by the advanced techniques discussed in Yan and Roy 2014 (2014), to handle the merging of segmented objects over time series. Our approach begins by extracting planted fields from each single-date image using ASAMPS with optimised prompt points (as shown in Figure 6(a-b)), then converting the extracted planted fields for each date into a vector map respectively. After that, a vector overlay operation is performed between two planted field vector maps obtained at two dates, and then the Maximum Intersection Ratio (MIR) is calculated for any two adjacent planted fields that came from two dates. Here, the MIR is defined as:

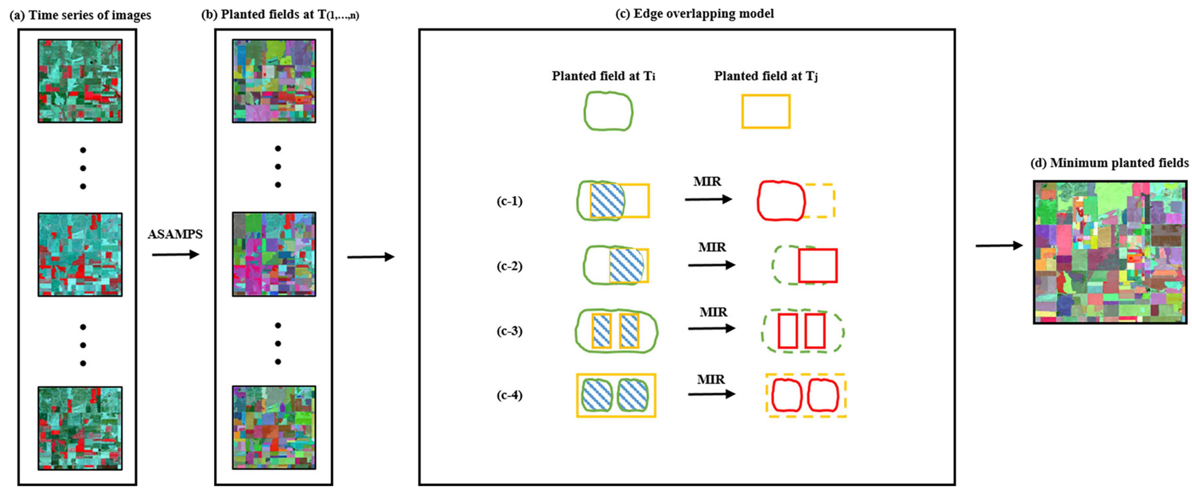

Figure 6. Schematic of the minimum planted area extraction by the edge overlapping module. (a) remote sensing imagery and (b) planted fields extracted by ASAMPS at different time points. (c) The edge-overlapping process, T_i and T_j represent different time points, the blue stripes indicate the overlapping area, and the red fields represent the planted fields retained in the final result.

$$MIR(f_i, f_j) = Max\left(\frac{area(f_i \cap f_j)}{area(f_i)}, \frac{area(f_i \cap f_j)}{area(f_j)}\right) \tag{6}$$

where, f_i and f_j denoted as two planted fields at different time points that need to be analysed, $area(f_i \cap f_j)$ is the overlapping area between the two planted fields and *area* (*) is the individual area of each planted field. *Max* is an operation of taking larger value. MIR values ranged from 0 to 1, with a larger value indicating that the area of one planted field is largely contained by the area of another (Figure 6(c)), and thus it is a smaller planted field (Figure 6(c)). We keep the smaller planted field and remove the larger one. By applying the same process to all the planted field vector maps in turn, we finally obtain the minimum planted fields of a growing season. This process can help to alleviate the problems that are caused by geometric registration errors to a certain extent. It should be noted that any omission, commission, over-segmentation and under-segmentation existing in the results of planted field segmentation from any single-date image can accumulate to the final result, so this module is an optional module only for the users who really need it.

2.4. Evaluation metrics

To evaluate the performance of our proposed method in extracting planted fields, this study employed three commonly used instance-based segmentation accuracy metrics (Lin et al. 2014; He et al. 2017), including the F_1-score, mean Average Precision (mAP), and Mean Intersection over Union (mIoU).

The F_1-score provides a balanced measure of the method's accuracy in segmentation, which is particularly useful for unbalanced datasets because it considers the impact of false positives and false negatives equally (He et al. 2017). It is first calculated for each instance (Equation 7), and then the average F1 score across all instances is calculated.

$$F_1 - \text{score} = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{7}$$

where Precision and Recall can be defined as $Precision = \frac{TP}{TP + FP}$, and $Recall = \frac{TP}{TP + FN}$. They are derived from a confusion matrix with four elements of "true positive" *(TP)*, "false positive" *(FP)*, "false negative" *(FN)*, and "true negative" *(TN)*.

The mean Average Precision (mAP) is a commonly used metric to evaluate model precision at different IoU thresholds (Lin et al. 2014). The mAP calculates the IoU between predicted instances and ground truth instances and computes the precision across multiple thresholds (e.g. IoU of 0.7, 0.75,..., 0.95), providing a comprehensive evaluation of the model's performance in instance segmentation tasks. First, the average precision (AP) at each IoU threshold can be calculated, typically using Equation 8.

$$AP = \sum_{n} (Recall_{\mathrm{n}} - Recall_{\mathrm{n}-1}) \cdot Precision_{\mathrm{n}} \tag{8}$$

where $Recall_{\mathrm{n}}$ and $Recall_{\mathrm{n}-1}$ are the adjacent Recall values, and $Precision_{\mathrm{n}}$ is the corresponding precision value. Then, the AP values across all IoU thresholds are averaged to obtain the mAP (Equation 9).

$$\mathrm{m}AP = \frac{1}{N} \sum_{i=1}^{\mathrm{N}} AP_i \tag{9}$$

where N is the number of IoU thresholds and AP_i is the AP value at the i-th IoU threshold. A higher mAP value indicates better performance in detecting instances of planted fields.

The Mean Intersection over Union (mIoU) is a standard metric for evaluating the accuracy of instance-level segmentation tasks. It calculates the ratio of the intersection over union between each predicted field instance and the corresponding ground truth instance (Equation 5) (He et al. 2017), and then averages these IoU values over all instances to obtain mIoU. This provides a comprehensive assessment of the model's performance, with higher values indicating more accurate segmentation results.

3. Study areas and experiments

3.1. Study areas and data

To test the performance of ASAMPS method, six typical agricultural areas were selected, four in China and two in the United States (Figure 7). The four areas in China (Songyuan City, Funan City, Rui'an City and Chengdu City) were chosen to test the performance of the ASAMPS method under various landscape complexities (Figure 7). The Songyuan City, located in the temperate monsoon climate zone with flat topography, shows the most homogeneous crop type (corn in summer) and the largest planted field size among the four studied areas. In contrast, Chengdu city, located in the subtropical humid monsoon climate zone with complex hilly terrain, presents diverse crop types and the most fragmented planted fields among the four study areas. Characteristics of Funan City and Rui'an City are between the above two study areas. We collected GF-2 PMS images acquired between April 2018 and September 2022 (Table 1), covering an area of approximately 30 km × 30 km of each study area. The Gream-Schmidt adaptive (GSA) method was used to fuse multispectral bands (blue, green, red, and near-infra-red bands with 4 m spatial resolution) with the corresponding panchromatic band image (1 m spatial resolution) to produce multispectral images with 1 m spatial resolution.

In the USA, a study area (30 km × 30 km) located in Pueblo County, a typical agricultural region of Colorado state (Figure 7), was selected to test the performance of ASAMPS method with images of different spatial resolutions, including Planet (3 m spatial resolution), Sentinel-2 (10 m spatial resolution), and Landsat 8 OLI (30 m spatial resolution) images acquired on October 17, 2020 (Table 1). Another study area (30 km × 30 km) located in Mason county, the southern part of the state of Iowa (Figure 7) within the corn-soybean belt with relatively homogeneous crop types and larger planted field sizes. It was used to test the performance of ASAMPS method for multi-temporal images.

3.2. Experiment I: the performance under various landscape complexities

This experiment aimed to test the generalisation and performance of ASAMPS method under different landscape complexities (Figure 7(a)). The state-of-the-art IAFRes model (Waldner and Diakogiannis 2020), which is a deep-learning-based method with a multi-task semantic segmentation architecture, was chosen as the benchmark method. In addition, ViT Adaptor, known for its effectiveness in instance segmentation tasks in remote sensing imagery (He et al. 2017; Freitas et al. 2024), was also chosen as a

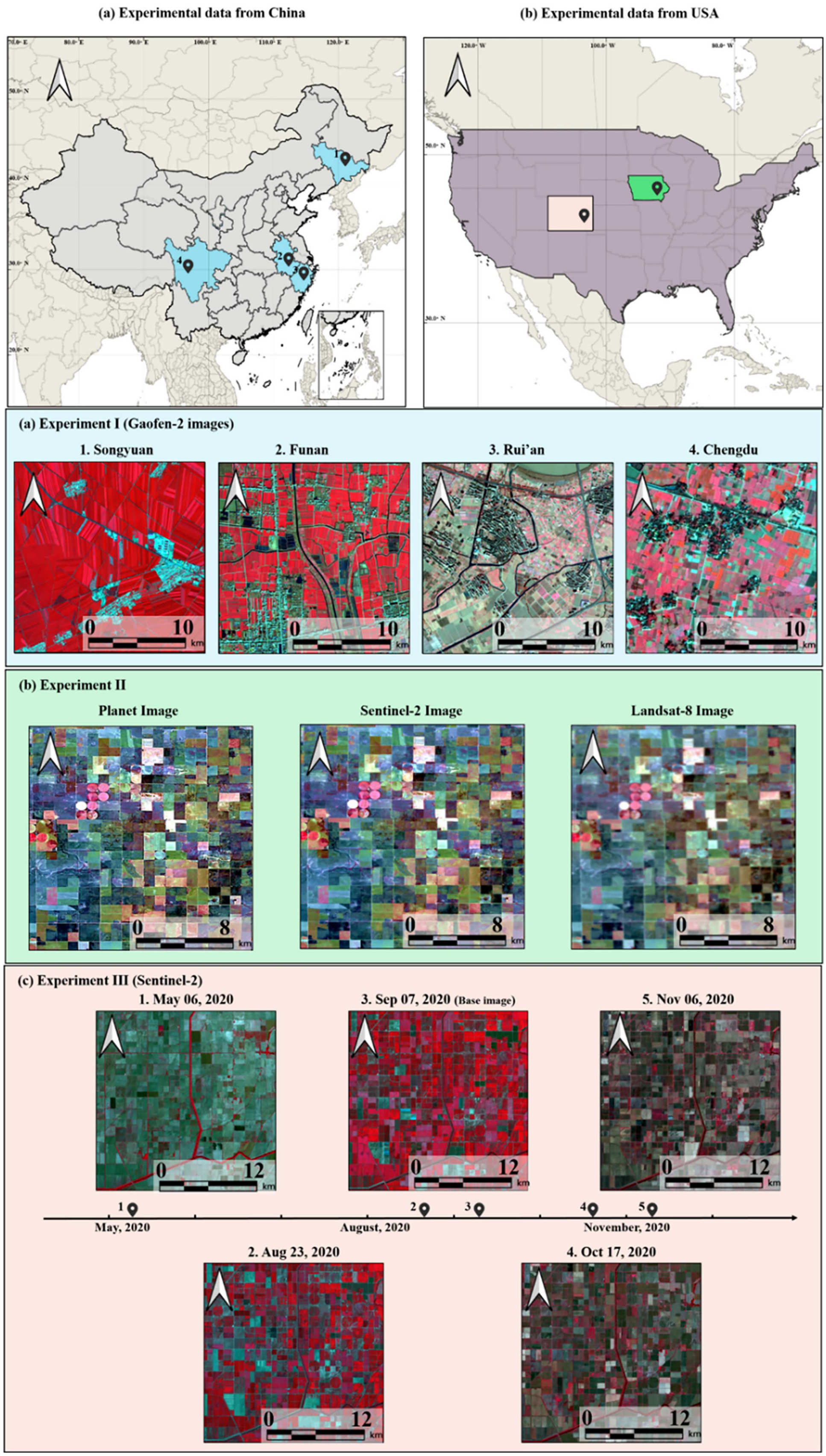

Figure 7. The locations of the study areas from China (a) and USA (b) and the images used in three experiments, shown in RGB composite with NIR, red and green bands.

Table 1. Description of the study sites and data used.

	Site	Data source	Location	Main crops	Acquisition date
Ex-1	Songyuan	Gaofen-2	45°2'N,125°52'E	Maize and soybean	2022/9/11
	Funan		32°36'N,115°24'E	Maize, wheat, paddy and vegetables	2020/7/29
	Rui'an		27°46'N,120°34'E	Paddy, wheat, canola and vegetables	2021/1/17
	Chengdu		31°2'N,104°0'E	Paddy, wheat, canola and vegetables	2018/4/24
Ex-2	Mason	Planet Sentinel-2 Landsat-8	40°21'N,107°26'W	Paddy, maize, wheat, canola and peanuts	2020/10/17
Ex-3	Pueblo	Sentinel-2	46°26'N,97°23'W	Maize, soybean and potato	2020/5/06–2020/11/06

Note: the area of each image in use is 30 × 30 km.

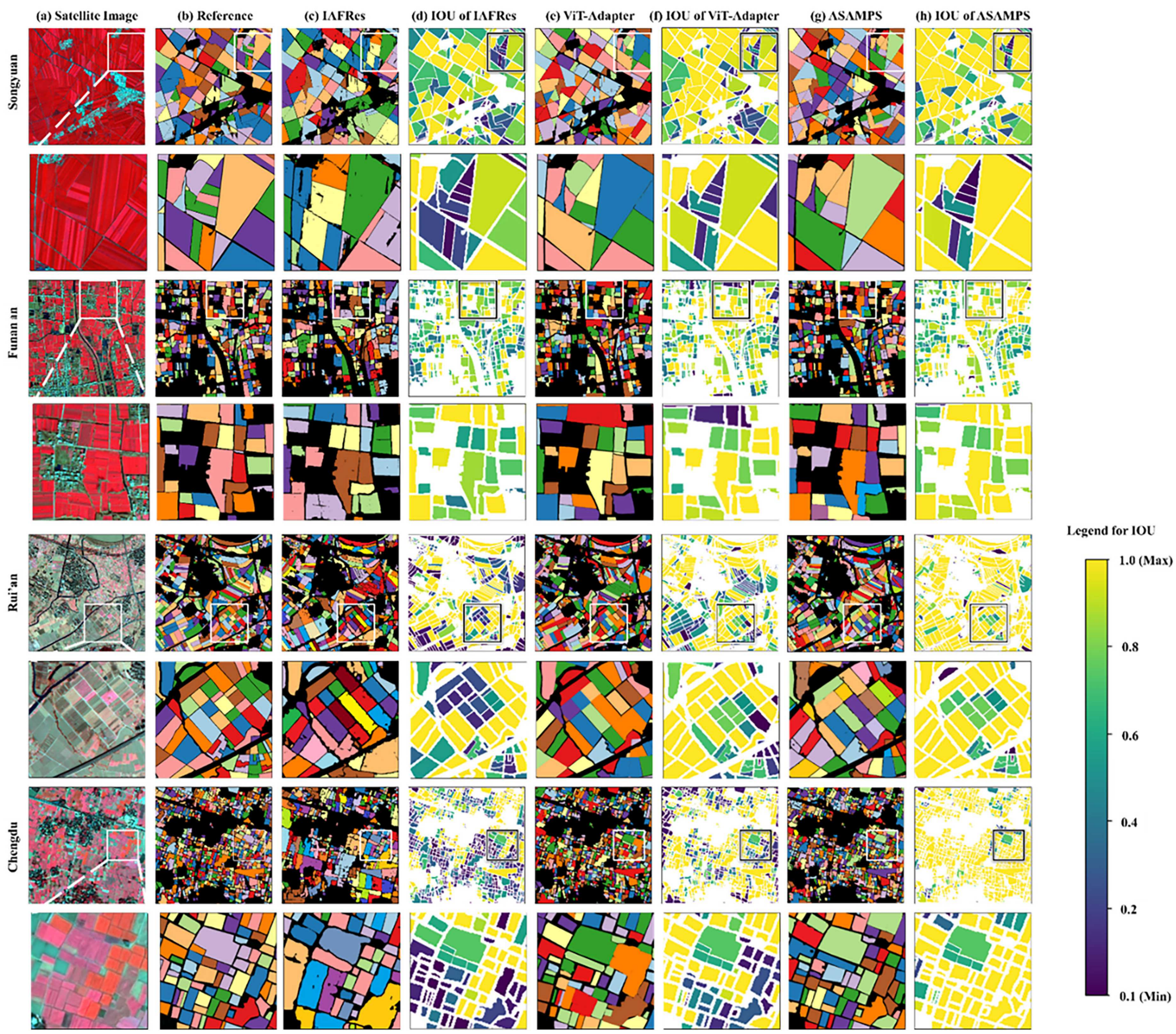

Figure 8. Visual comparison of planted field extraction results from Experiment I (performance under various landscape complexities). Rows from top to bottom: results for the four study sites (Songyuan, Funan, Rui'an, Chengdu). Columns from left to right: (a) Input false-colour composite GF-2 images (NIR, Red, Green bands); (b) Manually annotated reference labels, where each colour represents a distinct planted field; (c) Results generated by the IAFRes model; (d) IoU spatial distribution map corresponding to (c), where blue indicates low IoU (poor match) and yellow indicates high IoU (good match); (e) Results generated by the ViT Adaptor model; (f) IoU spatial distribution map corresponding to (e); (g) Results generated by the proposed ASAMPS model; (h) IoU spatial distribution map corresponding to (g).

representative instance segmentation method for comparison with our method. First, we manually labelled the planted fields of the different study area data with full coverage using the GF-2 data (see Figure 8(b)). These labels were meticulously cross-checked and refined to ensure high accuracy and consistency, forming a reliable ground truth dataset to validate the performance of the models used in the study.

Second, some image patches (192 patches of 512×512 pixels for each site) were clipped from the GF-2 data around each site and then the planted fields were manually labelled. These data were used to train the IAFRes and ViT Adaptor models. The training and validation data were spatially independent. ASAMPS did not undergo any training and was used directly on the GF-2 data of the study areas. The false-colour images were used in the experiments. They are a composite of the near-infra-red, green, and red bands. The reason for using false-colour images can be found in the Discussion section.

3.3. Experiment II: the performance with images of different spatial resolutions

This experiment aims to evaluate the cross-scale generalisation ability of ASAMPS on different images. The experiment used Landsat 8 OLI (30 m), Sentinel-2 (10 m), and Planet (3 m) images acquired at the Pueblo site on October 17, 2020 (Figure 7(b)). The full-coverage planted fields were manually labelled using the Planet images (see Figure 9(b)), which were used as ground truth data to validate the performance of the models used in the study. Meanwhile, 192 image patches (512×512 pixels) of planted fields around the study area were also manually labelled using the Planet data, which were used to train the IAFRes and ViT Adaptor models. The Landsat 8 OLI, Sentinel-2, and Planet images were then segmented using these models trained with planted field labels at a spatial resolution of 3 m, and the performance was validated with planted field labels at a spatial resolution of 3 m. ASAMPS was not trained in any way and

Figure 9. Visual comparison of planted field extraction results from Experiment II (performance across different spatial resolutions). Rows from top to bottom: results for the three satellite sensors (Planet, Sentinel-2, Landsat 8 OLI). Columns from left to right: (a) Input false-colour composite images (NIR, Red, Green bands); (b) Manually annotated reference labels using the Planet image as base, where each colour represents a distinct planted field; (c) Results generated by the IAFRes model; (d) IoU spatial distribution map corresponding to (c), where blue indicates low IoU and yellow indicates high IoU; (e) Results generated by the ViT Adaptor model; (f) IoU spatial distribution map corresponding to (e); (g) Results generated by the proposed ASAMPS model; (h) IoU spatial distribution map corresponding to (g).

was used directly on the Planet, Landsat 8 OLI, and Sentinel-2 images. The false-colour images were used in the experiment.

3.4. Experiment III: the performance of ASAMPS method from multi-temporal imagery

To evaluate the temporal generalisation of ASAMPS, we conducted this experiment on a cropland area in Mason county, Iowa. The dataset for this area consisted of five Sentinel-2 images (10 m) from May to November 2020, covering the entire summer growing season (see Figure 7(c)). Full-coverage planted fields were manually labelled using each of the 5 Sentinel-2 images. We selected the Sentinel-2 image acquired on September 7, 2020, as the base image and trained the IAFRes and ViT Adaptor models using the labelled planted fields (192 image patches of 512×512 pixels from the areas surrounding the study site, acquired on September 7, 2020). These trained IAFRes and ViT Adaptor models were then used to segment Sentinel-2 images from other dates. ASAMPS was applied directly to all Sentinel-2 images. Moreover, we generated the minimum planted fields in the summer growing season by overlaying all planted fields in the five Sentinel-2 images extracted by ASAMPS. The false-colour images were used in the experiments.

4. Results

4.1. Experiment I: the performance under various landscape complexities

Figure 8 shows the segmentation results of the three methods used on four different areas. When the IAFRes and ViT Adaptor models were trained using the local training data, all three methods exhibited good agreement with the reference images, but the performance of ASAMPS and ViT Adaptor was significantly better than that of IAFRes. This indicates that the performance of ASAMPS on zero-shot tasks is comparable to, or even superior to, the specifically trained IAFRes and ViT Adaptor models. Moreover, as the complexity of the landscape increased from Songyuan to Chengdu, IAFRes and ViT Adaptor produced more errors. ASAMPS maintained a similar trend but with fewer errors than IAFRes and ViT Adaptor (Figure 8(d, e, f)). The degradation in the performance of all three models may be due to inherent limitations in processing complex landscapes, where the accurate extraction of deep features typically becomes more difficult as the landscape complexity increases. More importantly, it was found that ASAMPS can segment more planted fields than IAFRes, and the extracted planted field boundaries are more topologically closed. This is because ASAMPS focuses more on the homogeneity of planted fields and can distinguish planted fields with soft edges, while IAFRes focuses more on hard edges (Waldner and Diakogiannis 2020). In addition, the extracted field instances by ViT Adaptor had overall systematic biases in boundary discrimination, with noticeable misclassifications and omissions at the boundaries. These boundary misclassifications further contributed to a marked drop in the IOU calculation, highlighting the challenges in accurately capturing smaller field boundaries with this method. Quantitative validation based on accuracy metrics (Table 2) was consistent with visual comparison results. All of these results demonstrate that ASAMPS is highly generalisable to areas of varying landscape complexity.

Table 2. Quantitative evaluations of planted field extraction results generated by IAFRes, ViT Adaptor and ASAMPS within four study areas. Mean is calculated as the average of each metric between different regions. Bold values indicate the most accurate results.

Site		Songyuan	Funan	Rui'An	Chengdu	(mean)
F*$_1$*-score	**IAFRes**	0.617	0.626	0.574	0.632	0.612
	ViT Adaptor	0.781	0.812	0.723	0.689	0.751
	ASAMPS	**0.849**	**0.827**	**0.808**	**0.752**	**0.809**
mAP	**IAFRes**	0.631	0.622	0.604	0.569	0.606
(IoU > 0.7)	**ViT Adaptor**	0.821	0.818	0.818	0.768	0.806
	ASAMPS	**0.852**	**0.840**	**0.826**	**0.796**	**0.828**
mIoU	**IAFRes**	0.609	0.587	0.601	0.565	0.591
	ViT Adaptor	0.811	0.798	0.746	0.801	0.789
	ASAMPS	**0.815**	**0.812**	**0.806**	**0.811**	**0.811**

4.2. Experiment II: the performance with images of different spatial resolutions

Figure 9 shows the segmentation results of the three methods used on images of different spatial resolutions obtained from the same experimental area. ASAMPS demonstrated strong cross-resolution generalisation capabilities (Figure 9(g) and Table 3), with F1-score, mAP, and mIoU (0.856, 0.877, and 0.842, respectively) for different resolution images, which were significantly higher than those of IAFRes (0.774, 0.769, and 0.770) and ViT Adaptor (0.848, 0.860, and 0.830). This can be attributed to the fact that ASAMPS was trained on the largest training dataset (SA-1B), consisting of 11 million different high-resolution images and 1.1 billion segmentation masks, ensuring the model's generalisability. In contrast, it was observed that IAFRes trained on the Planet image achieved acceptable planted field extraction results on the Planet image (see Figure 9(c-d) and Table 3). However, when the IAFRes model trained on the Planet image was directly applied to the Sentinel-2 and Landsat-8 images with different spatial resolutions of the same area, there was a noticeable decrease in extraction performance. The degraded performance of IAFRes is likely due to its strong dependence on the training data as a deep-learning-based method, meaning that the model can only learn the deep features contained in the Planet images. In contrast, the deep features in the Sentinel-2 and Landsat-8 images differ from those in the Planet images. The completeness of the training data and the significant difference in the amount of training data may be the main reasons for the performance discrepancy. Similarly, it was observed that ViT Adaptor trained on the Planet images also achieved comparable planted field extraction results to ASAMPS on the Planet images (see Table 3). However, when ViT Adaptor model was directly applied to the Sentinel-2 and Landsat-8 images, it produced significant omissions, leading to a rapid decline in extraction accuracy especially in mAP, and mIoU. The IoU spatial distribution of IAFRes and ViT adaptor (as shown in Figure 9(d) and (f)) notably varied with changes in spatial resolution, whereas ASAMPS, despite some fluctuations, consistently outperformed other methods in both field boundaries and field count accuracy. The significant differences in model structure and the amount of training data may be the main reason for the performance differences. These results highlight the advantage of ASAMPS in terms of transferability to images of different resolutions.

4.3. Experiment III: the performance of ASAMPS method from multi-temporal imagery

Figure 10 displays the performance differences of the three methods on images acquired at different dates in the same region. The performance of ASAMPS varied less over time and consistently achieved higher accuracy (F1 score, mAP, and mIoU values: 0.809, 0.833, and 0.798, respectively) than IAFRes and ViT Adaptor across different growth stages (Table 4). On the other hand, the IAFRes and ViT Adaptor models trained on the base date image (September 7, 2020) provided satisfactory planted field extraction results on September 7, 2020, with good agreement with the reference image. However, as the time interval between the image acquisition date and the base date increased (e.g. May 6, 2020 and November 6, 2020), the errors in the planted field extraction results of IAFRes increased (white boxes in Figure 10 and Table 4), and ViT Adaptor also showed some omissions at the field level. In contrast, ASAMPS maintained a high IOU across all dates, with values ranging from 0.759 to 0.843. These results demonstrate the strong temporal transferability and robustness of ASAMPS.

Table 3. Quantitative evaluations of planted field extraction results generated by IAFRes, ViT Adaptor and ASAMPS on images with different spatial resolutions. Mean is calculated as the average of each metric between different images. Bold values indicate the most accurate results.

Sensor		Planet	Sentinel-2	Landsat-8	(mean)
F_1-score	**IAFRes**	0.803	0.792	0.727	0.774
	ViT Adaptor	0.863	0.847	0.835	0.848
	ASAMPS	**0.871**	**0.850**	**0.846**	**0.856**
mAP	**IAFRes**	0.795	0.808	0.703	0.769
	ViT Adaptor	0.885	0.847	0.850	0.860
	ASAMPS	**0.895**	**0.858**	**0.861**	**0.877**
mIoU	**IAFRes**	0.837	0.775	0.698	0.770
	ViT Adaptor	0.852	0.827	0.812	0.830
	ASAMPS	**0.872**	**0.833**	**0.821**	**0.842**

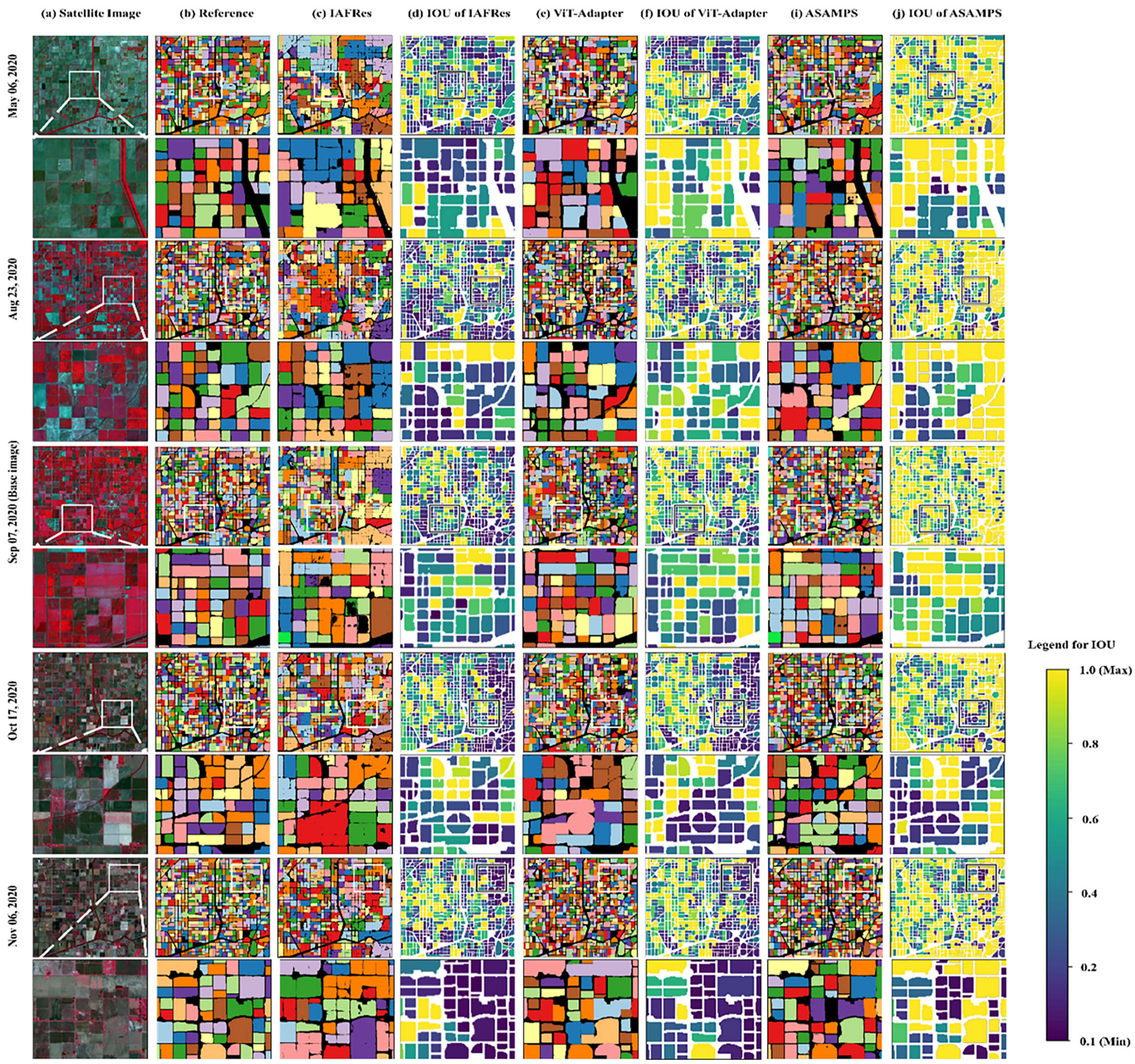

Figure 10. Visual comparison of planted field extraction results from Experiment III (performance on multi-temporal imagery). Rows from top to bottom: results for the five acquisition dates (May 6, Aug 23, Sep 7, Oct 17, Nov 6, 2020). Columns from left to right: (a) Input false-colour composite Sentinel-2 images (NIR, Red, Green bands); (b) Manually annotated reference labels for each date, where each colour represents a distinct planted field; (c) Results generated by the IAFRes model trained on the Sep 7 base date; (d) IoU spatial distribution map corresponding to (c), where blue indicates low IoU and yellow indicates high IoU; (e) Results generated by the ViT Adaptor model trained on the Sep 7 base date; (f) IoU spatial distribution map corresponding to (e); (g) Results generated by the proposed ASAMPS model; (h) IoU spatial distribution map corresponding to (g).

Table 4. Quantitative evaluations of planted field extraction results generated by IAFRes, ViT Adaptor and ASAMPS on images acquired at different dates. Mean is calculated as the average of each metric between different dates. Bold values indicate the most accurate results.

Date		6-May-20	23-Aug-20	7-Sep-20	17-Oct-20	6-Nov-20	(mean)
F_1-score	IAFRes	0.660	0.790	0.828	0.718	0.726	0.744
	ViT Adaptor	0.699	0.790	0.840	0.744	0.767	0.768
	ASAMPS	**0.727**	**0.812**	**0.846**	**0.817**	**0.841**	**0.809**
mAP	IAFRes	0.730	0.758	0.853	0.773	0.754	0.774
	ViT Adaptor	0.804	0.830	0.851	0.789	0.762	0.807
	ASAMPS	**0.833**	**0.841**	**0.86**	**0.803**	**0.829**	**0.833**
mIoU	IAFRes	0.732	0.791	0.833	0.721	0.676	0.751
	ViT Adaptor	0.767	0.820	0.831	0.742	0.708	0.774
	ASAMPS	**0.794**	**0.833**	**0.843**	**0.759**	**0.760**	**0.798**

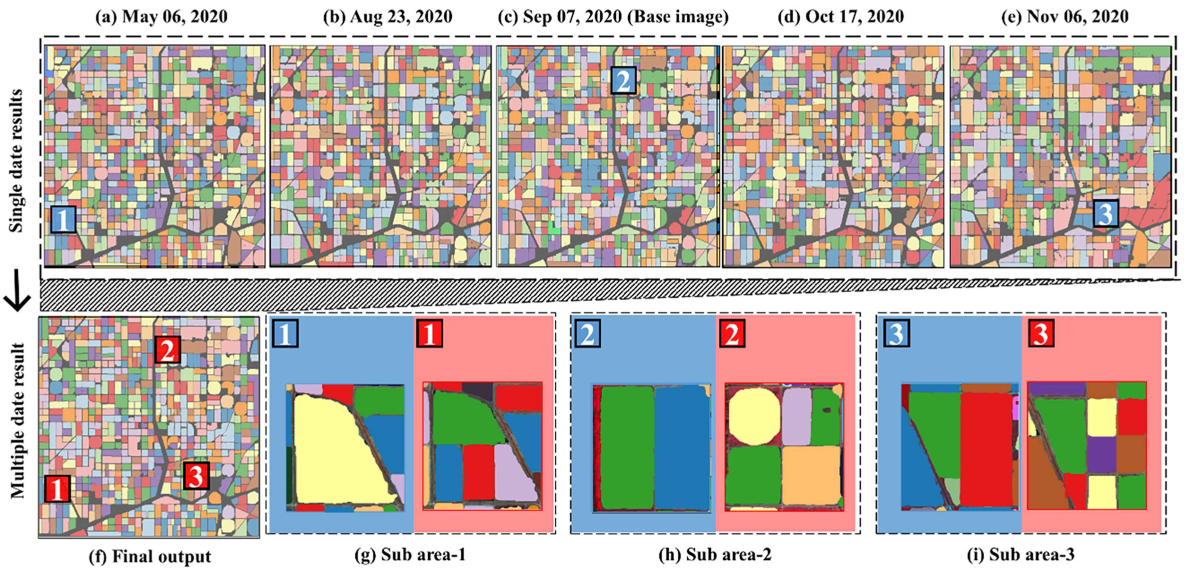

Figure 11. Synthesis of multi-temporal planted field extraction and identification of the minimum planted fields for an entire growing season. (a–e) Planted fields extracted by ASAMPS from Sentinel-2 images acquired on five different dates in 2020. (f) The final synthesised map of minimum planted fields for the entire growing season, obtained by applying the edge-overlapping module to the results from (a–e). Each colour represents a unique planted field persistent through the season. (g–i) Detailed views (corresponding to the numbered boxes in a–f) showcasing the dynamic changes and the overlay process. In these details, the planted fields extracted from individual dates are shown with a blue background, while the final, consolidated minimum planted fields retained for the growing season are depicted against a red background. The blue stripes in the schematic (c) represent the overlapping areas between fields from different date.

Figure 11 illustrates the planted field extraction results for a growing season (2nd row) from a synthesis of the temporal segmentation results previously obtained by ASAMPS model (1 st row). This was achieved through the edge overlapping module of ASAMPS. It is clear that ASAMPS preserves the spatial detail of planted fields at different spatial locations at different times. This is particularly evident in the areas within the boxes in Figure 11(g–i), which show the contrast between the spatial extent of the planted fields initially extracted at different time points and the final planted fields within a growing season. The final planted fields within a growing season further subdivided the distribution of planted fields over time, identifying minimum planted fields that were spatially separated by crop type or tillage practice.

5. Discussion

5.1. Ablation study on core modules

To verify the necessity and synergistic contribution of each core module in ASAMPS, ablation experiments were conducted based on Experiment III (multi-temporal imagery segmentation in Mason County, Iowa). The experimental settings (data sources, evaluation metrics, and computing environment) remained consistent with Experiment III, ensuring the reliability of comparative results.

Three ablation variants were constructed by sequentially removing each core module, with the complete ASAMPS model serving as the baseline: 1)Prompt Generation Module: Replace the task-driven Planted Index (PIE)-based prompt generation with SAM2's default grid sampling (5×5 grid) for prompt points, maintaining other modules unchanged. 2)Prompt Optimisation Module: Retain the PIE-generated prompt points but omit the dilation window-based redundancy removal and Non-Maximum Suppression (NMS) optimisation steps. 3) Edge-Overlapping Module: Use only the segmentation results of the base date (September 7, 2020) without fusing multi-temporal edge information, abandoning the Maximum Intersection Ratio (MIR)-based fusion strategy.

Quantitative results of the ablation experiments (mean values across five multi-temporal Sentinel-2 images) are presented in Table 5.

Table 5. Quantitative results of ablation experiments.

	Experiment III					
Model Variant	Prompt Generation	Prompt Optimisation	Edge-Overlapping	F1-score	mAP (IoU > 0.7)	mIoU
Ablation-1	x	o	o	0.687	0.702	0.675
Ablation-2	o	x	o	0.743	0.765	0.731
Ablation-3	o	o	x	0.771	0.794	0.762
Ablation-4	o	o	o	**0.809**	**0.833**	**0.798**

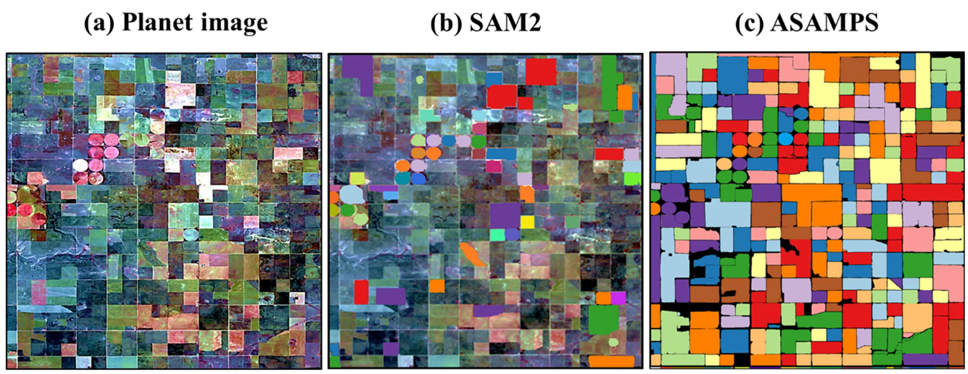

Figure 12. Visual comparison of planted field extraction on the same area obtained by SAM2 (b) and ASAMPS (c) models. The different colours within image (shown in (b) and (c)) represent different planted fields.

As shown in Table 5, removing any core module leads to a significant performance decline, confirming the indispensable role of each component. The Prompt Generation Module contributes the most to model performance. Its removal results in a 15.1% drop in F1-score and a 15.4% drop in mIoU, as SAM2's default grid sampling generates blind and invalid prompts (e.g. located at field edges or non-agricultural areas) that fail to guide accurate segmentation. The PIE-based task-driven design effectively addresses this issue by targeting the spectral homogeneity and edge gradient features of planted fields. Omitting the Prompt Optimisation Module causes an 8.2% decrease in F1-score. Redundant and misplaced prompt points lead to overlapping masks and missed fields, verifying that the dilation window + NMS strategy is critical for ensuring the spatial exclusivity and quality of prompt points. The Edge-Overlapping Module enhances model robustness in multi-temporal scenarios. Without it, the model cannot resolve boundary ambiguity caused by crop phenology changes, resulting in a 4.7% drop in F1-score. This module's MIR-based fusion strategy ensures the extraction of minimum stable planted fields across the growing season.

5.2. Linking SAM2 to planted field segmentation

SAM and SAM2 are designed for broader image segmentation tasks rather than specific segmentation tasks (Kirillov et al. 2023), which means that it focuses more on commonalities across tasks and may overlook some task specificities. Figure 12(b) shows an example of applying SAM2 directly to planted field extraction without adjusting any parameters. SAM2 can only extract a sparse number of planted fields with a large number of omission errors compared to ASAMPS (Figure 12(c)). The unsatisfactory performance of the direct SAM2 application underscores the necessity for domain-specific adaptations. Current endeavours to harness SAM in remote sensing primarily follow two paradigms: fine-tuning the SAM backbone (e.g. Qiao et al. 2025; Chen et al. 2023a) or employing learnable prompt generators (e.g. Zhang et al. 2023; Wang et al. 2025; Chen et al. 2023b). While effective, these methods often require retraining parts of SAM, incurring significant computational costs and, more critically, potentially compromising the model's inherently powerful zero-shot generalisation capability due to catastrophic forgetting or domain overfitting. In contrast, ASAMPS introduces a non-invasive, external adaptation paradigm. Our core innovation lies in the development of three auxiliary modules that operate entirely outside the frozen SAM2 architecture. This approach fundamentally differs from and offers unique advantages over existing methods:

First, ASAMPS developed a planted index (PIE) to generate prompt points by integrating the characteristics of planted fields: homogeneity and edge gradient features. In SAM and SAM2, prompt points are typically generated automatically by grid point sampling. This prompt point generation method is independent of the nature of the segmented objects. Setting the interval too small may result in large redundancy, and setting the interval too large may result in large omission. The randomness of the interval setting seriously affects the performance of SAM and SAM2 in a given task. In contrast, by combining the homogeneity and edge gradient features of planted fields, ASAMPS transforms the generation of prompt information from a random process into a more targeted and effective prompt selection strategy. This improvement significantly increases the efficiency and accuracy of ASAMPS in intensive planted field segmentation task.

Second, considering the significant exclusivity of planted fields, ASAMPS used the Non-Maximum Suppression (NMS) algorithm to remove the redundant points, in addition to using a dilation window instead of the traditional moving window. This makes each planted field to remain spatially independent and does not intersect other planted fields.

Third, an edge-overlapping module was developed to merge planted field vector maps obtained from single date image acquired at different dates. Since the spectral characteristics of different crops may be very similar in certain phenological periods, it is difficult to distinguish them in a single image just acquired at one phenological period, so merging the segmentation results from multiple temporal images can effectively distinguish different crops and obtain the minimum planted fields for a growing season. Previous deep learning algorithms (e.g. IAFRes and ViT Adaptor) rely heavily on training data. The challenge of generating multiple temporal training data has limited the temporal transferability of previous models, making it difficult to extract planted fields over time. As a result, this limits the ability to obtain the minimum planted fields for the entire growing season. The results of Experiment III further confirm this point and demonstrate the potential of ASAMPS in terms of temporal transferability and application, thus achieving minimum planted fields throughout a growing season. The above three points ensure that ASAMPS can fully combine the characteristics of the planted fields and thus show excellent applicability in extracting planted fields.

Finally, the computational efficiency of ASAMPS is also acceptable compared to existing methods under the same computing environment (i9-10900K (3.70 GHz) with 64 GB RAM, NVIDIA RTX 4080 with 32 GB VRAM). As shown in Table 6, ASAMPS requires 26 seconds to predict a single image for Experiment III, which is slightly longer than IAFRes but significantly shorter than ViT Adaptor. More importantly, ASAMPS does not require any training, which gives it a clear efficiency advantage over the other two models, which invest considerable time in training. This advantage is primarily due to the strong generalisation and transferability capabilities of the SAM2 model.

5.3. Linking SAM2 to remotely sensed imagery

The remote sensing community is actively exploring the application of SAM and SAM2 for specific domain applications. In particular, previous studies have tested the generalisation performance of SAM trained on natural images to high-resolution remote sensing imagery, i.e. segmenting objects such as solar panels, buildings, roads, and clouds, although applying SAM directly to remote sensing imagery was not as ideal as expected (Ji et al. 2023; Ren et al. 2023). One of the main reasons for this is that the characteristics of remotely sensed imagery have not been fully taken into account. There are significant differences between remote sensing imagery and natural images (Sun et al. 2023): natural images are usually acquired by standard cameras in the visible wavelengths (R, G, B bands) and are more focused on adaptation to the human visual system, while remote sensing imagery are acquired by high-altitude sensors that record not only visible light but also a wide range of spectral information (e.g. near-infra-red bands) and provide a

Table 6. The computation time of IAFRes, ViT Adaptor, and ASAMPS.

	Experiment III					
Model	Training Time (hrs)	Training GPU Mem(GB)	Inference Time (s)	Inference GPU Mem. (GB)	Model Size (MB)	Params (M)
IAFRes	0.16	5.2	21	3.1	108	27.5
ViT Adaptor	1.34	9.8	76	6.5	487	124.7
ASAMPS	N/A	N/A	26	4.3	1252	641

top-down perspective that is primarily focused on collecting spatial distribution information of the Earth's surface. These differences make it particularly important to consider the characteristics of remotely sensed imagery.

Since remotely sensed imagery provides richer spectral information than natural imagery, we tested the performance of ASAMPS on various multi-band composites. We selected Sentinel-2 satellite data in Experiment II and compared the near infra-red (NIR)-red-green band composite, the shortwave infra-red (SWIR-1)-NIR-red band composite, and the NDVI-red-green band composite to the red-green-blue band composite of the natural image. Here, SWIR-1, NIR, or NDVI was used instead of the blue band because these bands are more sensitive to vegetation than the blue band. As shown in Figure 13, the highest F_1-score value (0.872) is achieved by the NIR-red-green composites, followed by the SWIR1-NIR-red band (0.869), red-green-blue (0.861) and NDVI-red-green band (0.847) composites. This suggests that the use of more sensitive band composites has the potential to improve the performance of ASAMPS compared to the original band composite of the natural image. The reason for these results can be explained as follows: although SAM and SAM2 were trained on natural images, the large and diverse training data allowed it to focus more on the homogeneity and edge features of the images themselves (Kirillov et al. 2023). By incorporating bands that are more sensitive to vegetation, the edge features of vegetated areas are further enhanced, making it easier to distinguish between vegetated and non-vegetated areas, as well as between different crop types. In addition, since the NIR-red-green composite combines the most important spectral features of vegetation (strong chlorophyll reflection in the green band and strong absorption in the red band, multiple reflections from the vegetation canopy in the near-infra-red band), it is reasonable to expect that it will provide the best result compared to other composites. Given that NIR band data are more readily available from various remote sensing sensors, we recommend using the NIR-red-green band composite image in ASAMPS for vegetated field segmentation.

In addition, remote sensing imagery provide spatial distribution information about the Earth's surface, which can be represented abstractly by category information obtained by clustering methods using spectral and spatial features of the images. The clustering process helps ASAMPS to recognise the approximate spatial distribution of planted fields and thus helps to generate prompt points from the spatial distribution of planted fields. Accordingly, we recommend the use of clustered category information in ASAMPS, which is conducive to the utilisation of more spectral bands of remote sensing imagery and the spatial distribution features represented by the category information.

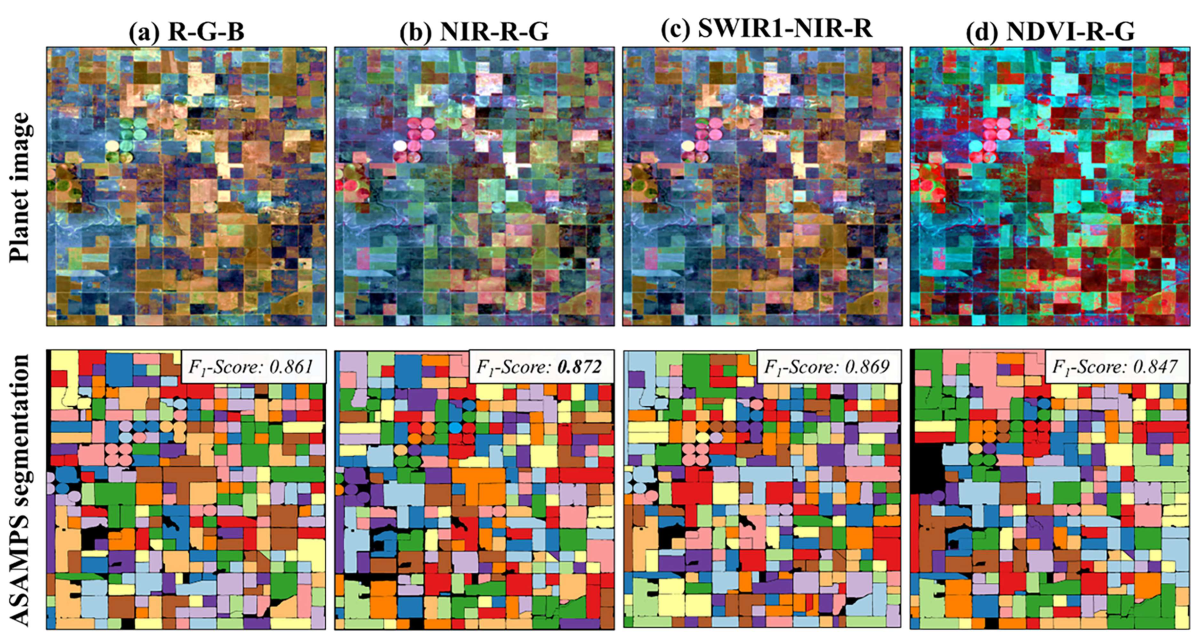

Figure 13. Visual comparison of planted field extraction by ASAMPS on the same image using different band combinations. The different colours within the image (shown in the bottom row) represent different planted fields.

5.4. Sensitivity analysis to planting index (PIE) threshold and window parameters

ASAMPS is a planted field extraction method based on prompt points, where the number of prompt points is crucial for the complete extraction of all planted fields. To achieve optimal planted field extraction results, it is advisable to define a suitable PIE threshold to adjust the number of these prompt points. In general, smaller PIE thresholds generate more prompt points, which may cause prompt point redundancy and consume more computational resources, while larger PIE thresholds generate fewer prompt points, which may cause a large number of planted field omissions. Accordingly, we investigated the performance of ASAMPS with decreasing PIE threshold using the Planet image of Experiment II (Figure 14). It is evident that as the threshold is decreased, the F_1-score of the extracted planted fields gradually increases, but the extraction time also continues to increase. The F_1-score for planted fields extraction peaks around a threshold of 0.85 (P_3 of Figure 14). Beyond this point, the F_1-score remains nearly stable. However, the extraction time for ASAMPS continues to increase as the threshold continues to decrease and the number of prompt points increases. Therefore, a threshold of around 0.85 is recommended for ASAMPS to

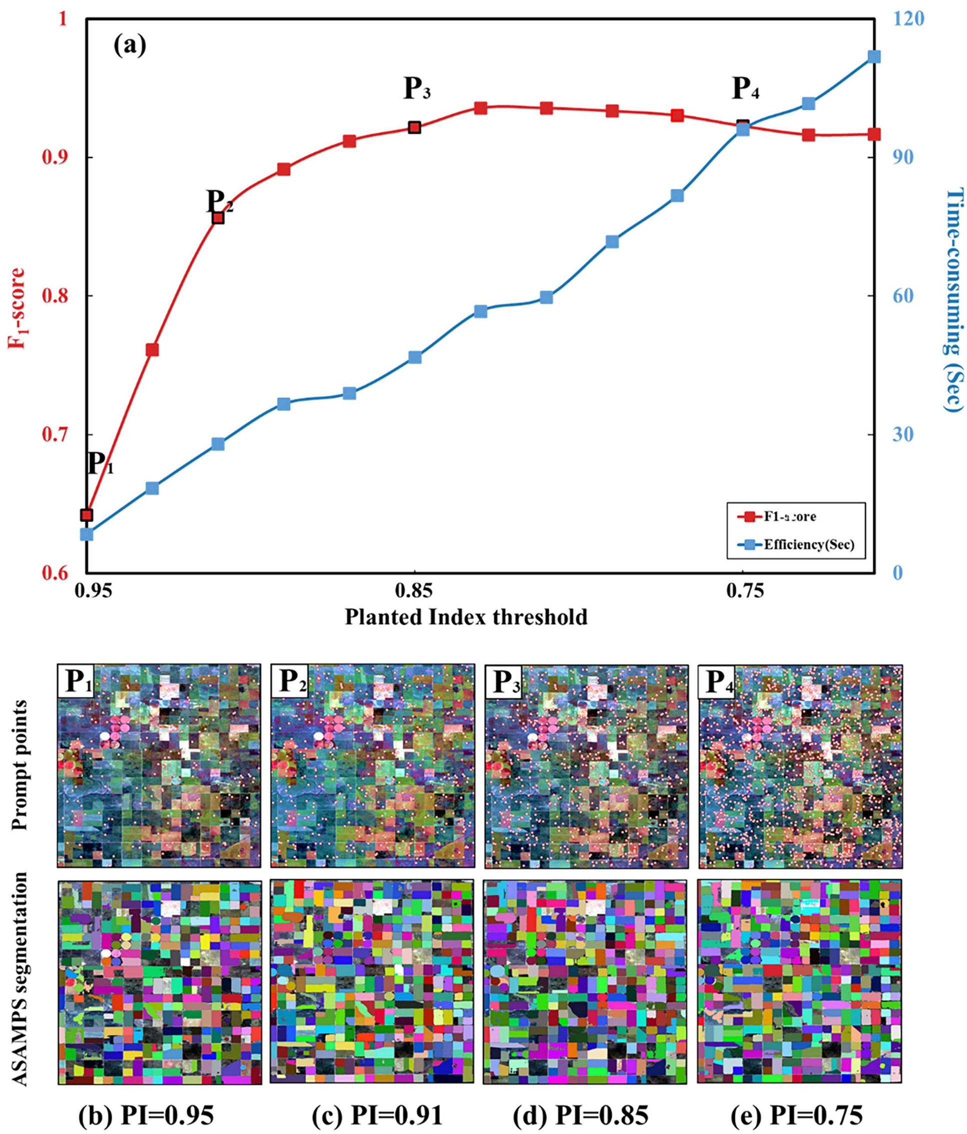

Figure 14. Effect of PIE threshold on planted field extraction by ASAMPS. Planted fields extraction results and corresponding prompt points (pink points in Figures) generated by PIE thresholds of 0.95, 0.91, 0.85, and 0.75 are shown in the bottom images as examples. The different colours shown in the bottom row represent different planted fields.

optimise the use of ASAMPS in practical scenarios, especially when dealing with time-sensitive tasks, as it maximises the F_1-score without significantly compromising the extraction speed.

Besides, selecting the appropriate parameters (window size and dilation rate) for the dilation window is critical to optimise prompt point positions (See Appendix II for definition of window size and dilation rate). The window size affects the number of prompt points that can be included in the dilation window; the dilation rate ensures that each selected prompt point maintains a certain spatial distance from the centre prompt point, which is conducive to reducing prompt point redundancy. To determine the appropriate window size and dilation rate for the dilation window, an experiment was conducted using the Planet image from Experiment II to investigate the effects of different window sizes and dilation rates on planted field segmentation accuracy and time cost (Figure 15). It can be seen that with a fixed dilation rate, as the window size increases, the F_1-score typically tends to first increase and then decrease. With a fixed window size, increasing the dilation rate results in a similar trend. In addition, a larger window size and a smaller dilation rate usually increases the computation time. By balancing the two sides of F_1-score and time consumption, a dilation window of window size 9 and dilation rate 2 is selected, which provides the highest extraction precision without significantly compromising the extraction speed.

5.5. Limitations and future prospects

Despite its robust performance, ASAMPS has certain limitations that warrant discussion and present directions for future research.

First, the model may occasionally miss very small planted fields (e.g. <100 m²). This is because extremely small areas often fail to generate interior points with sufficiently high Planted Index (PIE) values to surpass the threshold, as both gradient and homogeneity metrics become less reliable at very fine scales. Future work could explore multi-scale processing or incorporate super-resolution techniques to mitigate this issue. Second, the performance of ASAMPS is contingent upon the accuracy of the pre-masking land cover product (e.g. Dynamic World) used to isolate agricultural areas. Errors in this initial mask can propagate to the final result. Integrating uncertainty estimates from these products or developing a ensemble masking approach could enhance robustness. Third, the unsupervised clustering step within the PIE calculation requires defining the number of clusters, which can be subjective. While we provide general guidance, leveraging existing crop type maps or a small number of samples for supervised clustering could make this step more automated and

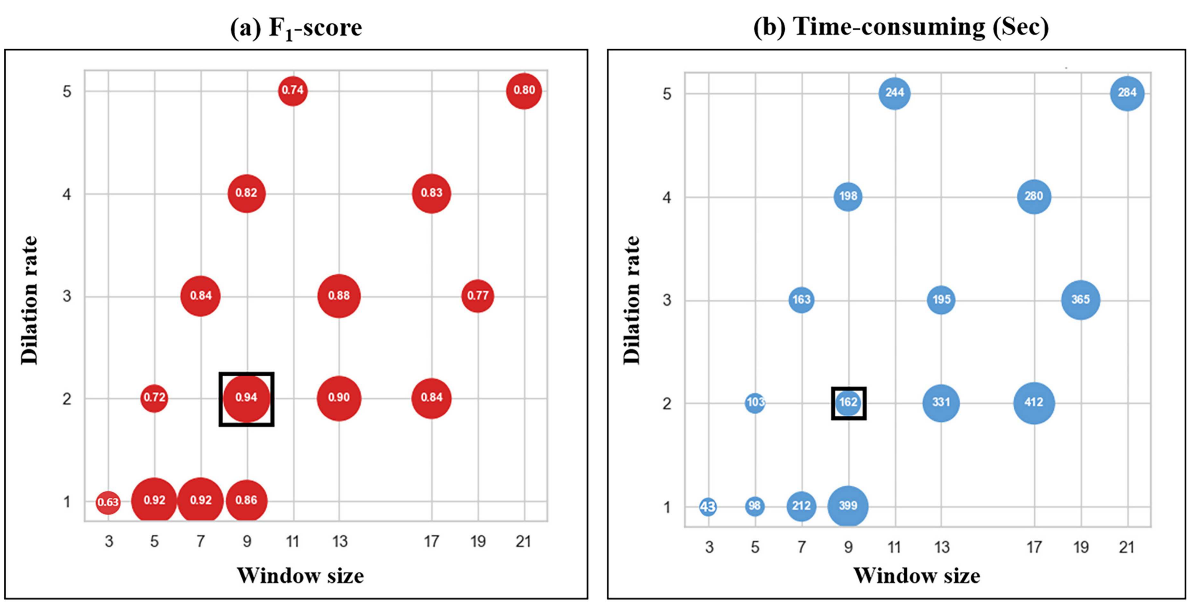

Figure 15. The effect of different window size and dilation rate of a dilation window on the accuracy (a) and efficiency (b) of ASAMPS. Larger circle represents higher F_1-score (a) and more time consuming (b). The black box indicates the optimal window size and dilation rate to balance F_1-score and time consumption.

accurate. Fourth, the efficacy of the edge-overlapping module is highly sensitive to the geometric precision of the input multi-temporal imagery. The primary purpose of this module is to identify the minimum stable areal unit (the planted field) present across a growing season by synthesising genuine changes in field boundaries (e.g. due to tillage, merging of plots, or crop rotation). Significant misregistration between images would indeed be amplified by the overlay process, leading to erroneous and fragmented polygons in the final output.

The primary application prospect of ASAMPS lies in large-scale, operational agricultural monitoring. Its ability to perform accurate, zero-shot field segmentation across space and time makes it ideally suited for generating consistent field parcels for crop type mapping, yield estimation, and sustainable agricultural practice management over vast areas. Beyond agriculture, the external adaptation strategy pioneered by ASAMPS holds significant promise for a wide array of remote sensing tasks. The core concept—using domain-specific features to generate optimal prompts for a frozen foundation model—could be adapted to other geospatial domains. For instance: In urban planning, a similar framework could be designed to generate prompts based on geometric features and texture patterns to segment individual buildings or infrastructure from high-resolution imagery. In environmental monitoring, features specific to forest stands (e.g. canopy texture, phenology) or water bodies could be encoded to develop prompt generators for forest species delineation or shoreline extraction. Future research will focus on abstracting this framework and developing task-specific indices to explore the transferability of the ASAMPS paradigm to these and other domains, such as natural hazard assessment.

6. Conclusion

In this study, we developed the ASAMPS model, a specialised adaptation of the Segment Anything Model 2 (SAM2), aimed at accurately extracting planted fields within an agricultural region. We designed a novel way to link SAM2 to domain-specific applications, i.e. to automatically generate prompts and optimise prompt locations completely outside of SAM2, based on the characteristics of the specific segmentation task and remotely sensed imagery. The advantage of this new approach is not only the complete preservation of the structure and strength of the SAM2 without additional retraining costs, but also the greater flexibility to redefine and optimise prompts for specific dense segmentation tasks. Our experiments in six different agricultural regions in China and the United States demonstrated the robustness of ASAMPS, which achieved comparable or better accuracy than state-of-the-art supervised methods. In particular, it performed well in complex landscapes and was adept at handling images with different spatial resolutions, such as those from the Planet, GF-2, Sentinel-2, and Landsat 8 OLI satellites. In addition, ASAMPS can effectively extract the minimum planted fields for a growing season with multi-temporal images. The current ASAMPS offers a novel framework for leveraging large foundation computer vision models in remote sensing applications and can be directly applied to various agricultural monitoring programmes, although it still has some limitations.

Author contributions

CRediT: **Shuaijun Liu:** Conceptualization, Data curation, Formal analysis, Investigation, Software, Visualization, Writing – original draft, Writing – review & editing; **Qi Dong:** Methodology, Writing – original draft; **Xuehong Chen:** Conceptualization, Data curation, Resources, Validation, Writing – original draft; **Xiuchun Dong:** Investigation, Methodology, Software, Validation; **Ping Huang:** Investigation, Resources; **Peng Yang:** Conceptualization, Methodology, Project administration, Resources; **Jin Chen:** Conceptualization, Resources, Software, Validation, Writing – original draft, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This research was funded by the National Natural Science Foundation for Young Scientists of China [grant number 42501460].

Data availability statement

The authors confirm that the data supporting the findings of this study are available within the article and its supplementary materials.

References

Aquino, A., M. E. Gegúndez-Arias, and D. Marín. 2010. "Detecting the Optic Disc Boundary in Digital Fundus Images Using Morphological, Edge Detection, and Feature Extraction Techniques." *IEEE Transactions on Medical Imaging* 29 (9): 1860–1869. https://doi.org/10.1109/TMI.2010.2053042.

Ball, G. H., and Hall, D. J. 1695. ISODATA, a novel method of data analysis and pattern classification. Stanford Research Institute, Menlo Park, CA, Technical Report No. AD699643.

Belgiu, M., and L. Drăguţ. 2014. "Comparing Supervised and Unsupervised Multiresolution Segmentation Approaches for Extracting Buildings from Very High Resolution Imagery." *ISPRS Journal of Photogrammetry and Remote Sensing* 96: 67–75. https://doi.org/10.1016/j.isprsjprs.2014.07.002.

Belgiu, M., and O. Csillik. 2018. "Sentinel-2 Cropland Mapping Using Pixel-Based and Object-Based Time-Weighted Dynamic Time Warping Analysis." *Remote Sensing of Environment* 204: 509–523. https://doi.org/10.1016/j.rse.2017.10.005.

Brown, C. F., S. P. Brumby, B. Guzder-Williams, T. Birch, S. B. Hyde, J. Mazzariello, W. Czerwinski, V. J. Pasquarella, R. Haertel, and S. Ilyushchenko. 2022. "Dynamic World, Near Real-Time Global 10 M Land Use Land Cover Mapping." *Scientific Data* 9 (1): 251. https://doi.org/10.1038/s41597-022-01307-4.

Cai, Z. and N. Vasconcelos. 2018. Cascade mask R-CNN. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 7107–7115.

Chen, Z., Duan, Y., Wang, W., He, J., Lu, T., Dai, J., and Qiao, Y. 2022. Vision Transformer Adapter for Dense Predictions. arXiv preprint arXiv:2205.08534. https://doi.org/10.48550/arXiv.2205.08534.

Chen, K., C. Liu, H. Chen, H. Zhang, W. Li, Z. Zou, and Z. Shi. 2023a. "RSPrompter: Learning to Prompt for Remote Sensing Instance Segmentation Based on Visual Foundation Model." *IEEE Transactions on Geoscience and Remote Sensing* 62: 1–17. https://doi.org/10.1109/TGRS.2024.3356074.

Chen, T., Zhu, L., Ding, C., Cao, R., Zhang, S., Wang, Y., Li, Z., Sun, L., Mao, P., and Zang, Y. 2023b. SAM Fails to Segment Anything? – SAM-Adapter: Adapting SAM in Underperformed Scenes: Camouflage, Shadow, and More. arXiv preprint, arXiv:2304.09148. https://doi.org/10.48550/arXiv.2304.09148.

Da Costa, J. P., F. Michelet, C. Germain, O. Lavialle, and G. Grenier. 2007. "Delineation of Vine Parcels By Segmentation of High Resolution Remote Sensed Images." *Precision Agriculture* 8 (1): 95–110. https://doi.org/10.1007/s11119-007-9031-3.

Freitas, P., G. Vieira, J. Canário, W. F. Vincent, P. Pina, and C. Mora. 2024. "A Trained Mask R-CNN Model over PlanetScope Imagery for Very-High Resolution Surface Water Mapping in Boreal Forest-Tundra." *Remote Sensing of Environment* 304: 114047. https://doi.org/10.1016/j.rse.2024.114047.

Gao, X., L. Liu, and H. Gong. 2020. "MMUU-Net: A Robust and Effective Network for Farmland Segmentation of Satellite Imagery." *Journal of Physics: Conference Series* 1651 (1): 012189. https://doi.org/10.1088/1742-6596/1651/1/012189.

García-Pedrero, A., C. Gonzalo-Martín, and M. Lillo-Saavedra. 2017. "A Machine Learning Approach for Agricultural Parcel Delineation through Agglomerative Segmentation." *International Journal of Remote Sensing* 38 (6): 1809–1819. https://doi.org/10.1080/01431161.2016.1278312.

He, K., G. Gkioxari, P. Dollár, and R. Girshick. 2017. Mask R-Cnn In *Proceedings of the IEEE International Conference on Computer Vision, 2017, 2961–2969*. 2980–2988. https://doi.org/10.1109/ICCV.2017.322.

Huang, H., Q. Huang, and P. Krahenbuhl. 2018. Domain Transfer through Deep Activation Matching In *Computer Vision – ECCV 2018. Lecture Notes in Computer Science*V. Ferrari, M. Hebert, C. Sminchisescu and Y. Weiss, eds. vol. 11210. 590–605, Cham: Springer. https://doi.org/10.1007/978-3-030-01270-0_36.

Huang, Z., L. Huang, Y. Gong, C. Huang, and X. Wang. 2019. Mask Scoring R-CNN. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR),* 6409–6418.

Ji, W., Li, J., Bi, Q., Li, W., and Cheng, L. 2023. Segment anything is not always perfect: An investigation of SAM on different real-world applications. arXiv preprint, arXiv:2304.05750 https://doi.org/10.48550/arXiv.2304.05750.

Karra, K., C. Kontgis, Z. Statman-Weil, J. C. Mazzariello, M. Mathis, and S. P. Brumby. 2021. "Global Land Use/Land Cover with Sentinel 2 and Deep Learning." *In: 2021 IEEE International Geoscience and Remote Sensing Symposium IGARSS*, 4704–4707. https://doi.org/10.1109/IGARSS47720.2021.9553499.

Khan, H., S. Iftikhar, R. Ward, S. Davy, and J. G. Breslin. 2025. "BiAgriNet: Binarized knowledge-distilled network for real-time semantic segmentation in agriculture." *IEEE Access* 13: 192374–192390. https://doi.org/10.1109/ACCESS.2025.3629617.

Kirillov, A., E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, and W.-Y. Lo. 2023. Segment Anything In *Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, 4015–4026*. 3992–4003. https://doi.org/10.1109/ICCV51070.2023.00371.

Lebourgeois, V., S. Dupuy, É. Vintrou, M. Ameline, S. Butler, and A. Bégué. 2017. "A Combined Random Forest and OBIA Classification Scheme for Mapping Smallholder Agriculture at Different Nomenclature Levels Using Multisource Data (Simulated Sentinel-2 Time Series, VHRS and Dem)." *Remote Sensing* 9 (3): 259. https://doi.org/10.3390/rs9030259.

Li, Y., H. Qi, J. Dai, X. Ji, and Y. Wei. 2017. Fully Convolutional Instance-aware Semantic Segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR),* 4476–4484.

Lin, T.-Y., M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick. 2014. Microsoft Coco: Common Objects in Context In *Computer Vision – ECCV 2014. Lecture Notes in Computer Science*D. Fleet, T. Pajdla, B. Schiele and T. Tuytelaars, eds. vol 8693. 740–755, Cham: Springer. https://doi.org/10.1007/978-3-319-10602-1_48.

Liu, S., J. Liu, X. Tan, X. Chen, and J. Chen. 2024. "A Hybrid Spatiotemporal Fusion Method for High Spatial Resolution Imagery: Fusion of Gaofen-1 and Sentinel-2 over Agricultural Landscapes." *Journal of Remote Sensing* 4: 0159. https://doi.org/10.34133/remotesensing.0159.

Liu, J., E. Pattey, J. R. Miller, H. McNairn, A. Smith, and B. Hu. 2010. "Estimating Crop Stresses, Aboveground Dry Biomass and Yield of Corn Using Multi-Temporal Optical Data Combined with a Radiation Use Efficiency Model." *Remote Sensing of Environment* 114 (6): 1167–1177. https://doi.org/10.1016/j.rse.2010.01.004.

Liu, S., L. Liu, F. Xu, J. Chen, Y. Yuan, and X. Chen. 2022. "A Deep Learning Method for Individual Arable Field (IAF) Extraction with Cross-Domain Adversarial Capability." *Computers and Electronics in Agriculture* 203: 107473. https://doi.org/10.1016/j.compag.2022.107473.

Long, J., E. Shelhamer, and T. Darrell. 2015. Fully Convolutional Networks for Semantic Segmentation In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2015, 3431–3440*. https://doi.org/10.1109/CVPR.2015.7298965.

Ma, L., Y. Liu, X. Zhang, Y. Ye, G. Yin, and B. A. Johnson. 2019. "Deep Learning in Remote Sensing Applications: A Meta-Analysis and Review." *ISPRS Journal of Photogrammetry and Remote Sensing* 152: 166–177. https://doi.org/10.1016/j.isprsjprs.2019.04.015.

Matton, N., G. Canto, F. Waldner, S. Valero, D. Morin, J. Inglada, M. Arias, S. Bontemps, B. Koetz, and P. Defourny. 2015. "An Automated Method for Annual Cropland Mapping Along the Season for Various Globally-Distributed Agrosystems Using High Spatial and Temporal Resolution Time Series." *Remote Sensing* 7 (10): 13208–13232. https://doi.org/10.3390/rs71013208.

Meyer, L., F. Lemarchand, and P. Sidiropoulos. 2020. "A Deep Learning Architecture for Batch-Mode Fully Automated Field Boundary Detection." *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences* XLIII-B3-2020: 1009–1016. https://doi.org/10.5194/isprs-archives-XLIII-B3-2020-1009-2020.

Osco, L. P., Q. Wu, E. L. de Lemos, W. N. Gonçalves, A. P. M. Ramos, J. Li, and J. Marcato. 2023. "The Segment Anything Model (SAM) for Remote Sensing Applications: From Zero to One Shot." *International Journal of Applied Earth Observation and Geoinformation* 124: 103540. https://doi.org/10.1016/j.jag.2023.103540.

Persello, C., V. A. Tolpekin, J. R. Bergado, and R. A. de By. 2019. "Delineation of Agricultural Fields in Smallholder Farms from Satellite Images Using Fully Convolutional Networks and Combinatorial Grouping." *Remote Sensing of Environment* 231: 111253. https://doi.org/10.1016/j.rse.2019.111253.

Qiao, Y., B. Zhong, B. Du, H. Cai, J. Jiang, Q. Liu, A. Yang, J. Wu, and X. Wang. 2025. "SAM Enhanced Semantic Segmentation for Remote Sensing Imagery Without Additional Training." *IEEE Transactions on Geoscience and Remote Sensing* 63: 1–16. https://doi.org/10.1109/TGRS.2025.3531930.

Rabbi, J., N. Ray, M. Schubert, S. Chowdhury, and D. Chao. 2020. "Small-Object Detection in Remote Sensing Images with End-To-End Edge-Enhanced GAN and Object Detector Network." *Remote Sensing* 12 (9): 1432. https://doi.org/10.3390/rs12091432.

Ravi, N., Gabeur, V., Hu, Y. T., Hu, R., Ryali, C., Ma, T., and Feichtenhofer, C. 2024. Sam 2: Segment anything in images and videos. arXiv preprint, arXiv:2408.00714. https://doi.org/10.48550/arXiv.2408.00714.

Ren, Y., X. Yang, Z. Wang, G. Yu, Y. Liu, X. Liu, D. Meng, Q. Zhang, and G. Yu. 2023. "Segment Anything Model (SAM) Assisted Remote Sensing Supervision for Mariculture—Using Liaoning Province, China as an Example." *Remote Sensing* 15 (24): 5781. https://doi.org/10.3390/rs15245781.

Rothe, R., M. Guillaumin, and L. Van Gool. 2014. Non-maximum suppression for object detection by passing messages between windows[C]. In *Asian conference on computer vision,* 290–306, Cham: Springer International Publishing.

Segl, K., and H. Kaufmann. 2001. "Detection of Small Objects from High-Resolution Panchromatic Satellite Imagery Based on Supervised Image Segmentation." *IEEE Transactions on Geoscience and Remote Sensing* 39 (9): 2080–2083. https://doi.org/10.1109/36.951105.

Su, Y. Y., S. Wei, S. Liu, J. Liang, C. Wang, J. Shi, and X. Zhang. 2020. "HQ-ISNet: High-Quality Instance Segmentation for Remote Sensing Imagery." *Remote Sensing* 12(6): 989. https://doi.org/10.3390/rs12060989.

Sun, X., P. Wang, W. Lu, Z. Zhu, X. Lu, Q. He, J. Li, et al. 2023. "RingMo: A Remote Sensing Foundation Model with Masked Image Modeling." *IEEE Transactions on Geoscience and Remote Sensing* 61: 1–22. https://doi.org/10.1109/TGRS.2022.3194732.

Tian, C., X. Chen, J. Chen, R. Cao, and S. Liu. 2024. "FieldSeg-DA2.0: Further Enhancing the Spatiotemporal Transferability of an Individual Arable Field (IAF) Extraction Network Using Multisource Remote Sensing and Land Cover Data." *Computers and Electronics in Agriculture* 222: 109050. https://doi.org/10.1016/j.compag.2024.109050.

Verrelst, J., J. P. Rivera, G. Leonenko, L. Alonso, and J. Moreno. 2014. "Optimizing LUT-Based RTM Inversion for Semiautomatic Mapping of Crop Biophysical Parameters from Sentinel-2 and -3 Data: Role of Cost Functions." *IEEE Transactions on Geoscience and Remote Sensing* 52 (1): 257–269. https://doi.org/10.1109/TGRS.2013.2238242.

Wagner, M. P., and N. Oppelt. 2020. "Extracting Agricultural Fields from Remote Sensing Imagery Using Graph-Based Growing Contours." *Remote Sensing* 12 (7): 1205. https://doi.org/10.3390/rs12071205.

Waldner, F., and F. I. Diakogiannis. 2020. "Deep Learning on Edge: Extracting Field Boundaries from Satellite Images with a Convolutional Neural Network." *Remote Sensing of Environment* 245: 111741. https://doi.org/10.1016/j.rse.2020.111741.

Waldner, F., F. I. Diakogiannis, K. Batchelor, M. Ciccotosto-Camp, E. Cooper-Williams, C. Herrmann, G. Mata, and A. Toovey. 2021. "Detect, Consolidate, Delineate: Scalable Mapping of Field Boundaries Using Satellite Images." *Remote Sensing* 13 (21): 2197. https://doi.org/10.3390/rs13212197.

Wang, J., X. Zhang, X. Ma, W. Yu, and P. Ghamisi. 2025. "Auto-Prompting SAM for Weakly Supervised Landslide Extraction." *IEEE Geoscience and Remote Sensing Letters* 22(3567600). https://doi.org/10.48550/arXiv.2501.13426.

Xi, J. and J.-Z. Zhang. 2012. Edge Detection from Remote Sensing Images Based on Canny Operator and Hough Transform In *Advances in Computer Science and Engineering. Advances in Intelligent and Soft Computing,* Lin S. and Huang X., eds. vol 141. 807–814, Berlin: Springer. https://doi.org/10.1007/978-3-642-27948-5_108.

Yan, L., and D. P. Roy. 2014. "Automated Crop Field Extraction from Multi-Temporal Web Enabled Landsat Data." *Remote Sensing of Environment* 144: 42–64. https://doi.org/10.1016/j.rse.2014.01.006.

Yang, G., Q. Zhang, and G. Zhang. 2020. "EANet: Edge-Aware Network for the Extraction of Buildings from Aerial Images." *Remote Sensing* 12 (13): 2161. https://doi.org/10.3390/rs12132161.

Yoo, D., N. Kim, S. Park, and S. Paek. 2016. Pixel-level domain transfer. In *European Conference on Computer Vision,* 517–532, Cham: Springer International Publishing.

Zanaga, D., Van De Kerchove, R., Daems, D., De Keersmaecker, W., Brockmann, C., Kirches, G., Wevers, J., Cartus, O., Santoro, M., and Fritz, S. 2022. ESA WorldCover 10 m 2021 v200. https://doi.org/10.5281/zenodo.7254221.

Zhang, J., Zhou, Z., Mai, G., Mu, L., Hu, M., and Li, S. 2023. Text2Seg: Remote Sensing Image Semantic Segmentation via Text-Guided Visual Foundation Models. arXiv preprint, arXiv:2304.10597. https://doi.org/10.48550/arXiv.2304.10597.

Zhu, X., E. H. Helmer, F. Gao, D. Liu, J. Chen, and M. A. Lefsky. 2016. "A Flexible Spatiotemporal Method for Fusing Satellite Images with Different Resolutions." *Remote Sensing of Environment* 172: 165–177. https://doi.org/10.1016/j.rse.2015.11.016.